Arithmetic Positivity and Absolute Hodge Classes: Why Hodge–Riemann Positivity Does Not Strengthen Deligne's Absoluteness

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We introduce the concept of *arithmetic positivity* for the Hodge Conjecture: a rational (p,p)-class $\alpha \in H^{2p}(X,\mathbb{Q})$ on a smooth projective variety $X/\mathbb{C}$ is called arithmetically positive if it simultaneously satisfies Hodge–Riemann positivity under all polarizations, Deligne's absoluteness under all automorphisms of $\mathbb{C}$, and Lefschetz compatibility. We prove that algebraic classes are arithmetically positive and record limited functoriality tests: finite pullback is controlled on the pullback ample subcone and becomes a full AP statement only under an explicit ample-cone comparison hypothesis; Künneth products are treated only as primitive product-polarization sign tests unless one invokes the later identity AP = AbsHodge; finite pushforward remains conjectural.

The central observation is a **boundary result**: the arithmetically positive cone coincides exactly with Deligne's absolute Hodge classes ($\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)$), because the Hodge–Riemann bilinear relations automatically guarantee the positivity conditions for any class of type (p,p). This is a well-known consequence of the Hodge–Riemann relations; the contribution of this paper is to make the saturation boundary explicit and to delineate what Hodge–Riemann-based positivity methods cannot achieve. The natural attempt to strengthen Deligne's absoluteness through Hodge–Riemann positivity therefore produces no new constraints, and the Arithmetic Positivity Conjecture reduces to Deligne's question (1982): "Are all absolute Hodge classes algebraic?" Furthermore, the obstructions (O1) and (O3) in the No-Go formulation are vacuous for rational (p,p)-classes; only (O2), failure of absoluteness, is effective. We introduce the Galois–Hodge–Riemann spectrum as a visualization of the absoluteness test, illustrate the theoretical predictions numerically across selected test configurations, and discuss controlled test-family directions for finding genuinely stronger conditions beyond the Hodge–Riemann relations.

This paper is an expository contribution documenting a structural limitation; it does not contain a new algebraicity theorem beyond the observation that AP = AbsHodge.

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CONTENTS

I. Introduction 3
A. The Hodge Conjecture 3
B. Limits of Construction-Only Strategies 3
C. The Positivity Alternative 3
D. Outline 4

II. Background 4
A. Hodge–Riemann Bilinear Form 4
B. Absolute Hodge Classes 4

III. Arithmetic Positivity 4
A. Definition 4
B. The Arithmetic Positivity Conjecture 5

IV. Algebraic Classes Are Arithmetically Positive 5

V. Functoriality of Arithmetic Positivity 6

VI. The No-Go Theorem 8
A. The Unconditional Statement 9

VII. Abelian Varieties 9

VIII. The Precise Obstruction 9
A. The Galois–Hodge–Riemann Form 9
B. Why the Obstruction Is Natural 11

IX. Discussion 11
A. The Remaining Gap 11
B. Relation to Deligne's Program 12
C. Connection to the Broader Program 13
D. Limitations 13
E. Outlook 13
F. Directions for Stronger Positivity 14
1. p-Adic Slope Constraints 14
2. Arakelov Positivity 14
3. Tannakian Perspective 14
4. Prismatic and Categorical Approaches 15
Threshold desideratum: beyond Deligne 15
5. Geometric and o-Minimal Approaches 16
G. Beyond Bilinear Positivity: Nonlinear and Averaged Conditions 17
H. Numerical Illustration 17
I. Controlled Test-Family Transfer Beyond AP = AbsHodge 17
Towards AP′: Provenance Defects 18
Towards AP″: Conditional Orbit-Rigidity 18
Towards AP‴: Uniformity and Anti-Advice 19
Scale and Multi-Erosion Guardrails 19
Result Classification 20

X. Conclusion 20

Funding Information 21

References 21

I. INTRODUCTION

A. The Hodge Conjecture

Let X be a smooth projective variety of dimension n over $\mathbb{C}$. The Hodge decomposition

$$H^k(X,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X) \tag{1}$$

expresses the k-th singular cohomology as a direct sum of Dolbeault cohomology groups. A class $\alpha \in H^{2p}(X,\mathbb{Q})$ is called a *rational Hodge class* if its image in $H^{2p}(X,\mathbb{C})$ lies entirely in $H^{p,p}(X)$. Every algebraic cycle Z of codimension p defines a rational Hodge class $\mathrm{cl}(Z) \in H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ via the cycle class map.

Conjecture I.1 (Hodge, 1950 [19])**.** *Every rational Hodge class on a smooth projective variety over $\mathbb{C}$ is a $\mathbb{Q}$-linear combination of classes of algebraic cycles.*

Despite over seven decades of effort, the conjecture remains open in general. It is known to hold for:

- Divisors ($p = 1$): by the Lefschetz $(1,1)$-theorem.
- Abelian varieties of dimension ≤ 5: by work of Hazama [18], Moonen [21], and Markman [22].
- Products of elliptic curves and certain Fermat hypersurfaces.
- Uniruled varieties (for degree-2 classes).

B. Limits of Construction-Only Strategies

Many approaches to the Hodge Conjecture attempt to *construct* an algebraic cycle representing a given Hodge class. Such constructions are indispensable in known special cases, but they also reveal structural bottlenecks. This paper tests a complementary route: ask whether non-algebraic Hodge classes can be ruled out by a positivity obstruction.

The structural bottlenecks relevant for this route are:

1. Direct algebraization. Given a rational (p,p)-class, attempt to construct a subvariety whose fundamental class matches it. This fails because there is no canonical projection from Hodge cohomology to algebraic cycles, and the construction is not functorial under deformation.

2. Variations of Hodge structure. Use families X_t and track how Hodge classes vary. This fails because Hodge classes can "jump"—the condition of being type (p,p) is not open in the Zariski topology—and monodromy destroys naive stability arguments [25].

3. Motivic methods. Replace varieties by motives and hope that Hodge classes are motivic. This fails because the category of motives is not fully constructed: Grothendieck's standard conjectures [17] remain open, and the relationship between Hodge classes and absolute Hodge classes is unresolved [2].

4. Reduction mod p. Reduce to characteristic p, invoke the Tate conjecture, and lift back. This fails because the bridge Hodge $\Rightarrow$ Tate $\Rightarrow$ Hodge is not closed: Frobenius invariance does not imply Hodge type, and lifting algebraic cycles is not unique [9].

As we show below, the positivity alternative proposed in this paper also encounters a fundamental limitation: the natural positivity conditions reduce to Deligne's absoluteness. Nevertheless, the analysis delineates a precise boundary for this class of approaches.

Table I. Positivity properties and their structural role across arithmetic and geometric conjectures.

Conjecture	**Positivity Invariant**	**Structural Role / Source**
Weil conjectures	Frobenius eigenvalues ≥ 0	Weight purity (Deligne [10])
BSD (rank 1)	Néron–Tate height $\hat{h} \geq 0$	Heegner points (Gross–Zagier [16])
Tate (abelian var.)	Isogeny positivity	Faltings' finiteness theorem
Hodge Conjecture	**Galois–HR spectrum**	**Saturated** (AP = AbsHodge)

C. The Positivity Alternative

We observe that major breakthroughs in adjacent conjectures follow a different pattern (Table I): In each case, a positivity property of the relevant arithmetic object plays a structural role—though the proof techniques differ substantially (Deligne's proof of the Weil conjectures uses ℓ-adic cohomology and the Lefschetz fixed-point theorem; Gross–Zagier explicitly constructs a Heegner point and computes its height). The Hodge Conjecture lacks: *a positivity structure that forces algebraicity without constructing cycles.*

This paper introduces such a structure—*arithmetic positivity*—combining Hodge–Riemann positivity under all polarizations with Deligne's absoluteness under all automorphisms of $\mathbb{C}$. The central finding, however, is a boundary/no-go observation: the Hodge–Riemann bilinear relations automatically guarantee the positivity conditions for any class of type (p,p), so the arithmetically positive cone coincides with Deligne's absolute Hodge classes: $\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)$. The automatic saturation of Hodge–Riemann positivity for (p,p)-classes is well known to specialists (cf. Voisin [25], Chapter 6; Griffiths–Harris [15], Chapter 0–1); the purpose of this paper is to make the saturation boundary explicit and systematic within a unified positivity framework, and to clarify that no Hodge–Riemann-based strengthening of Deligne's absoluteness is possible. The Arithmetic Positivity Conjecture reduces to Deligne's question: "Are all absolute Hodge classes algebraic?"

We regard this boundary result as informative: it delineates a precise limit for positivity-based approaches to the Hodge Conjecture and identifies directions for genuinely stronger conditions beyond Hodge–Riemann.

D. Outline

Section II reviews the Hodge–Riemann bilinear form and absolute Hodge classes. Section III introduces arithmetic positivity. Section IV proves that algebraic classes are arithmetically positive. Section V records limited functoriality tests and their hypotheses. Section VI formulates the No-Go theorem. Section VII proves the equivalence for abelian varieties. Section VIII identifies the precise obstruction. Section IX discusses the remaining gap and the relationship to Deligne's program.

This paper is self-contained; all main results depend only on definitions and proofs within this document. Connections to the broader FST program are cited for context but are not logically required.

II. BACKGROUND

A. Hodge–Riemann Bilinear Form

Let X be a smooth projective variety of dimension n, equipped with a Kähler form ω defining an ample class $L = [\omega] \in H^{1,1}(X) \cap H^2(X, \mathbb{Z})$. The *Lefschetz operator* $\Lambda_L : H^k(X) \to H^{k+2}(X)$ is defined by $\Lambda_L(\alpha) = L \cup \alpha$. We use the convention $\Lambda_L(\alpha) = L \cup \alpha$; some authors reserve Λ for the dual Lefschetz operator, while denoting the cup-product operation simply by L or $L\wedge$ (Voisin [25]; Griffiths–Harris [15]).

Theorem II.1 (Hard Lefschetz)**.** *For $k \le n$, the map $\Lambda_L^{n-k} : H^k(X, \mathbb{Q}) \to H^{2n-k}(X, \mathbb{Q})$ is an isomorphism.*

A class $\alpha \in H^k(X, \mathbb{Q})$ with $k \le n$ is *primitive* (with respect to L) if $\Lambda_L^{n-k+1}(\alpha) = 0$. The *Hodge–Riemann bilinear form* on the primitive cohomology $P^k(X, \mathbb{Q}) \subset H^k(X, \mathbb{Q})$ is:

$$Q_L(\alpha, \beta) = \int_X \alpha \cup \beta \cup L^{n-k}\,. \tag{2}$$

Theorem II.2 (Hodge–Riemann bilinear relations)**.** *On the primitive subspace $P^{p,q}(X) = P^{p+q}(X) \cap H^{p,q}(X)$ with $p + q = k$, the Hermitian form*

$$h_L(\alpha, \beta) = i^{p-q}\,(-1)^{k(k-1)/2}\,Q_L(\alpha, \bar{\beta}) \tag{3}$$

is positive definite.

The Hodge–Riemann relations are the fundamental positivity result in Hodge theory. They provide definiteness *relative to a fixed polarization L*. A central theme of this paper is that algebraicity requires positivity not just for one L, but for *all* admissible polarizations and operations.

B. Absolute Hodge Classes

Deligne [11] introduced a refinement of Hodge classes. An automorphism $\sigma \in \mathrm{Aut}(\mathbb{C})$ sends a smooth projective variety $X/\mathbb{C}$ to the conjugate variety X^σ (same equations, coefficients twisted by σ). The comparison isomorphism provides:

$$H^{2p}_{\mathrm{dR}}(X/\mathbb{C}) \xrightarrow{\sim} H^{2p}_{\mathrm{dR}}(X^\sigma/\mathbb{C})\,, \tag{4}$$

mapping the de Rham class of α on X to a class α^σ on X^σ.

Definition II.3 (Deligne)**.** A rational Hodge class $\alpha \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$ is *absolute* (or *absolutely Hodge*) if for every $\sigma \in \mathrm{Aut}(\mathbb{C})$, the conjugate class α^σ is again a Hodge class on X^σ.

Theorem II.4 (Deligne [11])**.** *1. Every algebraic cycle class is absolute.*

2. On abelian varieties, every Hodge class is absolute.

Deligne's result shows that absoluteness is a *necessary* condition for algebraicity. The question is whether it is sufficient. Our concept of arithmetic positivity *attempts* to strengthen absoluteness by adding Hodge–Riemann positivity under all conjugations; as shown in Section IX, this attempt does not succeed (the additional conditions are automatic), but the analysis is informative.

III. ARITHMETIC POSITIVITY

We now introduce the central concept.

A. Definition

Let X be a smooth projective variety of dimension n over $\mathbb{C}$, and let $\mathrm{Amp}(X)$ denote the ample cone of X.

Definition III.1 (Arithmetic Positivity)**.** A rational Hodge class $\alpha \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$ is *arithmetically positive* if it satisfies the following three conditions simultaneously:

When using the direct primitive notation below, we work in the middle-or-lower range $2p \le n$; classes in complementary codimension are understood via Poincaré duality. Likewise, the Lefschetz-compatibility condition is only imposed for cup-powers that still have a directly defined primitive component in the middle-or-lower range.

(AP1) Universal Hodge–Riemann positivity. For every ample class $L \in \mathrm{Amp}(X)$, the primitive component α_0 of α (with respect to L) satisfies the *signed* Hodge–Riemann inequality:

$$(-1)^p\,Q_L(\alpha_0, \alpha_0) \ge 0\,, \tag{5}$$

where Q_L is the Hodge–Riemann form (2). The sign $(-1)^p$ is essential: the Hodge–Riemann bilinear relations (Theorem II.2) give $(-1)^p Q_L > 0$ on primitive (p,p)-classes, not $Q_L > 0$.

(AP2) Absolute stability. For every $\sigma \in \mathrm{Aut}(\mathbb{C})$:

(a) α^σ is a Hodge class on X^σ (i.e., α is absolute).

(b) The conjugated class α^σ satisfies (AP1) on X^σ with respect to every ample class $L^\sigma \in \mathrm{Amp}(X^\sigma)$.

(AP3) Lefschetz compatibility. For all $0 \le j \le \lfloor n/2 \rfloor - p$ and all ample L, let $\gamma_j = \Lambda_L^j \alpha \in H^{2(p+j)}(X)$ and let $\gamma_{j,0}$ denote its primitive component with respect to L in $H^{2(p+j)}(X)$. Then:

$$(-1)^{p+j}\, Q_L(\gamma_{j,0},\, \gamma_{j,0}) \ge 0\,. \tag{6}$$

The set of arithmetically positive classes in $H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ is denoted $\mathrm{AP}^p(X)$.

The full AP condition is *a priori* stronger than Deligne's absoluteness alone; we will show in Section 8 that the two notions in fact coincide.

Remark III.2. The conditions (AP1), (AP2b), and (AP3) are *not* independent of (AP2a). As shown in Proposition IX.1 below, the Hodge–Riemann bilinear relations automatically guarantee (AP1), (AP2b), and (AP3) for any class of type (p,p). The only substantive condition is therefore (AP2a): Deligne's absoluteness. In particular, $\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)$ (Remark IX.2). The definition retains all three conditions for expository clarity: they make explicit the *type* of positivity that algebraic classes enjoy, even though the Hodge–Riemann relations render them automatic for (p,p)-classes. Finding genuinely stronger conditions – beyond the Hodge–Riemann relations – that *do* constrain beyond absoluteness remains an open problem (see Section IX).

B. The Arithmetic Positivity Conjecture

Conjecture III.3 (Arithmetic Positivity Conjecture)**.** *Let X be a smooth projective variety over $\mathbb{C}$. Then:*

$$\mathrm{AP}^p(X) = \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})\,, \tag{7}$$

where $\mathrm{CH}^p(X)_{\mathbb{Q}}$ is the Chow group of codimension-p cycles with rational coefficients, and cl *is the cycle class map.*

Proposition III.4. *The following relations hold between the APC and the HC:*

1. *The Hodge Conjecture implies the Arithmetic Positivity Conjecture.*
2. *The "hard direction" of the APC ($\mathrm{AP}^p(X) \subseteq \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$) together with the statement that all rational Hodge classes are arithmetically positive ($H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X) \subseteq \mathrm{AP}^p(X)$) implies the HC.*
3. *Unconditionally:* $\mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}}) \subseteq \mathrm{AP}^p(X)$ *(Theorem IV.1 below).*

Proof. (1) If the HC holds, then every Hodge class is algebraic, and algebraic classes satisfy (AP1)–(AP3) (Theorem IV.1 below). Thus $\mathrm{AP}^p(X) \subseteq H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X) = \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$, and the reverse inclusion holds by Theorem IV.1, giving $\mathrm{AP}^p(X) = \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$.

(2) This requires two separate assumptions. *First*, suppose that the hard direction of the APC holds: $\mathrm{AP}^p(X) \subseteq \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$, i.e., every arithmetically positive class is algebraic. *Second*, suppose that every rational Hodge class is arithmetically positive: $H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X) \subseteq \mathrm{AP}^p(X)$. Under both assumptions, every rational Hodge class α lies in $\mathrm{AP}^p(X)$ (by the second assumption) and hence in $\mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$ (by the first assumption), so α is algebraic. This yields the HC.

Note that neither assumption alone suffices: the APC ($\mathrm{AP}^p = \mathrm{alg}^p$) shows that AP classes are algebraic, but without knowing that all Hodge classes are AP, one cannot conclude that all Hodge classes are algebraic.

(3) See Theorem IV.1. □

Remark III.5. The APC alone does not directly imply the HC. The APC asserts $\mathrm{AP}^p = \mathrm{alg}^p$, while the HC asserts $\mathrm{Hodge}^p = \mathrm{alg}^p$. Since $\mathrm{AP}^p \subseteq \mathrm{Hodge}^p$ (arithmetic positivity imposes additional conditions beyond being a Hodge class), the APC gives $\mathrm{alg}^p \subseteq \mathrm{Hodge}^p$ (trivial) but not $\mathrm{Hodge}^p \subseteq \mathrm{alg}^p$ without the additional knowledge that all Hodge classes are arithmetically positive. The non-trivial content thus splits into two parts: (i) show that non-AP Hodge classes do not exist (i.e., every Hodge class is AP), and (ii) show that AP implies algebraic (the hard direction of APC).

The non-trivial content of the APC is the "hard direction": $\mathrm{AP}^p(X) \subseteq \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$. This says that arithmetic positivity *forces* algebraicity.

IV. ALGEBRAIC CLASSES ARE ARITHMETICALLY POSITIVE

Theorem IV.1 (Easy Direction)**.** *Let X be a smooth projective variety over $\mathbb{C}$. Every class in the image of the cycle class map is arithmetically positive:*

$$\mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}}) \subseteq \mathrm{AP}^p(X)\,. \tag{8}$$

Proof. Let $\alpha = \mathrm{cl}(Z)$ for some algebraic cycle Z of codimension p with rational coefficients. We verify (AP1)–(AP3).

(AP1): Let α_0 be the L-primitive component of $\alpha = \mathrm{cl}(Z)$. Since α is algebraic, it lies in $H^{p,p}(X) \cap H^{2p}(X,\mathbb{Q})$, and its primitive component α_0 lies in $P^{p,p}(X) \cap H^{2p}(X,\mathbb{Q})$. By the Hodge–Riemann bilinear relations (Theorem II.2), the Hermitian form h_L is positive definite on $P^{p,p}(X)$. For a rational (p,p)-class, $\bar{\alpha}_0 = \alpha_0$ (complex conjugation acts as the identity on classes of type (p,p) with rational coefficients), $i^{p-p} = 1$, and $(-1)^{k(k-1)/2} = (-1)^{p(2p-1)}$ with $k = 2p$. Thus $h_L(\alpha_0,\alpha_0) = (-1)^{p(2p-1)}\, Q_L(\alpha_0,\alpha_0)$. Since

$(-1)^{p(2p-1)} = (-1)^{2p^2-p} = (-1)^p$ (as $(-1)^{2p^2} = 1$), the Hodge–Riemann relations give:

$$(-1)^p\, Q_L(\alpha_0, \alpha_0) > 0. \tag{9}$$

if $\alpha_0 \neq 0$. For $\alpha_0 = 0$ (i.e., when α has no primitive component), the inequality holds trivially. This argument uses only the Hodge–Riemann relations on $P^{p,p}(X)$ and does not require α_0 itself to be the class of an effective cycle.

(AP2): Part (a) is Deligne's theorem (Theorem II.4): algebraic classes are absolute. Part (b) follows because σ sends algebraic cycles on X to algebraic cycles on X^σ (algebraicity is a property of polynomial equations, which transform covariantly under σ), and algebraic classes on X^σ satisfy (AP1) by the argument above.

(AP3): Cup product with an ample $(1,1)$-class preserves Hodge type: for admissible j in Definition III.1, the class $\gamma_j = \Lambda_L^j \alpha = L^j \cup \alpha$ lies in $H^{p+j,p+j}(X)$. If L is integral or rational, this can be represented by intersecting Z with hyperplane-section classes; for a general real ample class the cycle-language should be read only after rational approximation, while the sign argument itself uses only Hodge type. The L-primitive component $\gamma_{j,0}$ lies in $P^{p+j,\,p+j}(X)$, and the Hodge–Riemann relations give $(-1)^{p+j} Q_L(\gamma_{j,0}, \gamma_{j,0}) \geq 0$, by the same argument as in (AP1). □

V. FUNCTORIALITY OF ARITHMETIC POSITIVITY

For a full No-Go program, one would want the arithmetically positive cone to be functorial under the standard operations of algebraic geometry. Table II summarizes the functoriality and stability status of arithmetically positive classes across fundamental algebraic geometric operations, delineating the exact hypotheses, proof mechanisms, and open boundaries. The results below record the limited functoriality that is actually controlled here.

Proposition V.1 (Pullback: Finite Morphisms)**.** *Let $f : Y \to X$ be a finite morphism of smooth projective varieties. If $\alpha \in \mathrm{AP}^p(X)$, then $f^*\alpha$ satisfies (AP1) for ample classes in the pullback subcone $f^*\mathrm{Amp}(X) \subset \mathrm{Amp}(Y)$, and satisfies (AP2)–(AP3) on that subcone. It is a full element of $\mathrm{AP}^p(Y)$ only under an explicit ample-cone comparison hypothesis, for example when the relevant positivity inequalities extend from $f^*\mathrm{Amp}(X)$ to all of $\mathrm{Amp}(Y)$ by density or by equality of the numerical ample cones.*

Proof. Let $f : Y \to X$ be finite of degree d.

(AP1): Since f is finite, f^*L_X is ample on Y for every ample $L_X \in \mathrm{Amp}(X)$. For the primitive component α_0 of α with respect to L_X, the projection formula gives:

$$\int_Y f^*\alpha_0 \cup f^*\bar{\alpha}_0 \cup (f^*L_X)^{n-2p} = d \int_X \alpha_0 \cup \bar{\alpha}_0 \cup L_X^{n-2p},$$

where $n = \dim X = \dim Y$. Since $d > 0$ and the right side satisfies $(-1)^p Q_{L_X}(\alpha_0, \alpha_0) \geq 0$ by assumption, we obtain $(-1)^p Q_{f^*L_X}(f^*\alpha_0, f^*\alpha_0) \geq 0$ on the pullback ample subcone. For a general ample class $L_Y \in \mathrm{Amp}(Y)$, no extension follows from finiteness alone: the primitive component of $f^*\alpha$ varies with L_Y, and the pullback subcone need not be dense in the ample cone of Y. If an additional ample-cone comparison hypothesis gives density or equality of the relevant cone, then continuity of the Hodge–Riemann form and of the Lefschetz decomposition extends the inequality to all ample classes.

(AP2): The pullback morphism commutes with σ-conjugation: $(f^*\alpha)^\sigma = (f^\sigma)^*(\alpha^\sigma)$, and $f^\sigma : Y^\sigma \to X^\sigma$ is again finite of the same degree. Since $\alpha^\sigma \in \mathrm{AP}^p(X^\sigma)$ by assumption, the result follows from the (AP1) argument applied to f^σ and α^σ.

(AP3): The Lefschetz operator satisfies $\Lambda^j_{f^*L} \circ f^* = f^* \circ \Lambda^j_L$ (compatibility of pullback with cup product). Therefore $\Lambda^j_{f^*L}(f^*\alpha) = f^*(\Lambda^j_L \alpha)$, and the positivity of the primitive component follows from the same degree-d argument applied to $\Lambda^j_L \alpha$. □

Remark V.2. The degree formula $Q_{f^*L_X}(f^*\alpha_0, f^*\alpha_0) = \deg(f) \cdot Q_{L_X}(\alpha_0, \alpha_0)$ used in the proof relies on f being a finite morphism (equivalently, generically finite with $\dim Y = \dim X$). For a generically finite morphism $f : Y \to X$ with $\dim Y = \dim X$ that is not finite, the projection formula still yields the degree relation on a dense open subset, but care is needed regarding the ramification locus; the statement of Proposition V.1 is therefore restricted to finite morphisms.

Remark V.3. For morphisms $f : Y \to X$ with $\dim Y > \dim X$, the pullback f^*L_X is not ample on Y, and the primitive decomposition on Y requires a relative Hodge–Riemann analysis. The functoriality of AP^p under general morphisms remains open.

Remark V.4 (Density caveat). The extension from the pullback subcone $f^*\mathrm{Amp}(X)$ to all of $\mathrm{Amp}(Y)$ requires an explicit ample-cone comparison. A useful sufficient condition is that the pullback subcone has dense image in the relevant numerical ample cone, or that the numerical ample cones agree after pullback. This may hold in Picard-number-one situations, but it fails in general. Without such a hypothesis, Proposition V.1 is only a subcone functoriality statement.

Conjecture V.5 (Pushforward: Finite Morphisms)**.** *Let $f : Y \to X$ be a finite morphism of smooth projective varieties (so $\dim Y = \dim X$). If $\alpha \in \mathrm{AP}^p(Y)$, then $f_*\alpha \in \mathrm{AP}^p(X)$.*

Remark V.6 (Evidence status for finite pushforward). The following heuristic supports Conjecture V.5; it is not a proof. The projection formula $f_* f^* = \deg(f) \cdot \mathrm{id}$ on $H^*(X, \mathbb{Q})$ implies that f_* is injective on the image of f^* (up to scalar). For $\alpha \in \mathrm{AP}^p(Y)$ and any ample $L_X \in \mathrm{Amp}(X)$, the class f^*L_X is ample on Y (since f is finite). The Hodge–Riemann form on X is related to that on Y

Table II. Functoriality and stability verification matrix for arithmetic positivity across algebraic operations.

Operation	Domain / Codomain	Positivity Preservation	Proof Mechanism	Hypotheses & Edge Cases
Finite Pullback f^*	$\mathrm{AP}^p(X) \to \mathrm{AP}^p(Y)$	Preserved on $f^*\mathrm{Amp}(X)$	Degree-d projection formula	Requires ample cone comparison / density
Künneth Product $\boxtimes$	$\mathrm{AP}^p(X) \times \mathrm{AP}^q(Y) \to \mathrm{AP}^{p+q}$	Product polarization sign	Binomial expansion (Lem. V.11)	Cross-terms open for mixed polarizations
Cup Product $\cup$	$\mathrm{AP}^p(X) \times \mathrm{AP}^q(X) \to \mathrm{AP}^{p+q}$	Preserved on (p,p)-types	Saturation Lemma V.12	Automatic via $(p+q,p+q)$-type
Galois Action σ	$\mathrm{AP}^p(X) \to \mathrm{AP}^p(X^\sigma)$	Covariant unitary isometry	Conjugation covariance	Fails on transcendental classes if non-Hodge
Finite Pushforward f_*	$\mathrm{AP}^{p+d}(Y) \to \mathrm{AP}^p(X)$	Conjectural / Unproven	Adjoint trace pairing	Requires Hard Lefschetz on non-finite base

via the adjunction $\int_X f_*\beta \cup \gamma = \int_Y \beta \cup f^*\gamma$. Combined with the degree formula, this gives evidence for the sign conditions for $f_*\alpha$. The absolute stability is expected to transfer because f^σ is again finite of the same degree. A complete proof requires verifying that the primitive decomposition on X is compatible with f_* applied to primitive classes on Y; this is not straightforward, since f_* does not preserve primitivity in general.

Remark V.7. For morphisms $f : Y \to X$ with $\dim Y > \dim X$, the pushforward f_* involves a Gysin-type argument relating the Hodge–Riemann form on Y to that on X via the Grothendieck–Riemann-Roch formula and the relative primitive decomposition. This analysis is not carried out here; the functoriality of AP^p under general pushforward remains open.

Proposition V.8 (Künneth Product: Primitive Product-Polarization Sign Test)**.** *Let $\alpha \in \mathrm{AP}^p(X)$ and $\beta \in \mathrm{AP}^q(Y)$ be primitive with respect to ample classes L_X and L_Y. Then the direct product-form calculation proves the (AP1) sign inequality for $\alpha \boxtimes \beta$ for all product polarizations $L = L_X \boxtimes 1 + 1 \boxtimes L_Y$, and conjugation is compatible with the exterior product. This proposition is only a product-polarization sign test; it does not by itself prove the full (AP3) condition for all Lefschetz powers on $X \times Y$.*

Remark V.9. The obstruction is not merely that non-primitive input classes are harder. Even when α and β are primitive, the classes $\Lambda_L^j(\alpha \boxtimes \beta)$ generally decompose into sums whose L-primitive projections on $X \times Y$ contain cross-terms. The sign formula below controls the primitive exterior product itself under product polarizations. Full AP-membership of $\alpha \boxtimes \beta$ is obtained in this paper only through the later saturation identity AP = AbsHodge, together with the standard fact that exterior products of absolute Hodge classes are absolute; it is not a separate Künneth AP theorem.

We first establish a key lemma on primitivity of exterior products.

Lemma V.10 (Primitivity of exterior products)**.** *Let X and Y be smooth projective varieties of dimensions n and m respectively, with ample classes $L_X \in \mathrm{Amp}(X)$ and $L_Y \in \mathrm{Amp}(Y)$. Set $L = L_X \boxtimes 1 + 1 \boxtimes L_Y \in \mathrm{Amp}(X \times Y)$. If $\alpha \in P^{2p}(X, \mathbb{Q})$ is primitive with respect to L_X and $\beta \in P^{2q}(Y, \mathbb{Q})$ is primitive with respect to L_Y, then $\alpha \boxtimes \beta$ is primitive with respect to L on $X \times Y$.*

Proof. Set $k = p + q$ and $N = n + m = \dim(X \times Y)$. We must show $L^{N-2k+1} \cup (\alpha \boxtimes \beta) = 0$. Since $L = L_X \boxtimes 1 + 1 \boxtimes L_Y$, the binomial expansion gives for any $j \geq 0$:

$$L^j \cup (\alpha \boxtimes \beta) = \sum_{a+b=j} \binom{j}{a} (L_X^a \cup \alpha) \boxtimes (L_Y^b \cup \beta).$$

Each summand vanishes unless both factors are non-zero. Since α is L_X-primitive, $L_X^{n-2p+1} \cup \alpha = 0$, so $L_X^a \cup \alpha = 0$ for $a \geq n-2p+1$. Similarly, $L_Y^b \cup \beta = 0$ for $b \geq m-2q+1$. A summand with $a + b = j$ is non-zero only if $a \leq n - 2p$ and $b \leq m - 2q$, which requires $j \leq (n-2p) + (m-2q) = N - 2k$. Therefore $L^j \cup (\alpha \boxtimes \beta) = 0$ for all $j \geq N - 2k + 1$, which is precisely the primitivity condition. □

Lemma V.11 (Sign compatibility for exterior products)**.** *Under the hypotheses of Lemma V.10, with $k = p + q$:*

$$(-1)^k Q_L(\alpha \boxtimes \beta,\, \alpha \boxtimes \beta) = \binom{N-2k}{n-2p} \big[(-1)^p Q_{L_X}(\alpha, \alpha)\big] \cdot \big[(-1)^q Q_{L_Y}(\beta, \beta)\big], \tag{10}$$

where $N = n + m$ and Q_L, Q_{L_X}, Q_{L_Y} are the Hodge–Riemann forms (2) on $X \times Y$, X, and Y respectively.

Proof. By Lemma V.10, $\alpha \boxtimes \beta$ is L-primitive, so the definition (2) gives directly:

$$Q_L(\alpha \boxtimes \beta,\, \alpha \boxtimes \beta) = \int_{X \times Y} (\alpha \boxtimes \beta)^2 \cup L^{N-2k}.$$

Expanding the power L^{N-2k} by the binomial theorem as in Lemma V.10, the unique non-vanishing term occurs at $a = n - 2p$ and $b = m - 2q$ (forced by the constraints $a \leq n-2p$, $b \leq m-2q$, and $a+b = N-2k = (n-2p)+(m-2q)$). By the Künneth formula for integration:

$$\int_{X \times Y} (\alpha \boxtimes \beta)^2 \cup L^{N-2k} = \binom{N-2k}{n-2p} \left(\int_X \alpha^2 \cup L_X^{n-2p}\right) \cdot \left(\int_Y \beta^2 \cup L_Y^{m-2q}\right).$$

By definition (2), the integrals on X and Y are precisely $Q_{L_X}(\alpha, \alpha)$ and $Q_{L_Y}(\beta, \beta)$. Therefore:

$$Q_L(\alpha \boxtimes \beta,\, \alpha \boxtimes \beta) = \binom{N-2k}{n-2p} Q_{L_X}(\alpha, \alpha) \cdot Q_{L_Y}(\beta, \beta).$$

Multiplying both sides by $(-1)^k = (-1)^{p+q} = (-1)^p \cdot (-1)^q$ yields equation (10). □

Proof of Proposition V.8. We verify the controlled (AP1) product-polarization statement for $\alpha \boxtimes \beta$ with $\alpha \in \mathrm{AP}^p(X)$ primitive and $\beta \in \mathrm{AP}^q(Y)$ primitive.

(AP1): Let $L = L_X \boxtimes 1 + 1 \boxtimes L_Y$ for ample classes $L_X \in \mathrm{Amp}(X)$, $L_Y \in \mathrm{Amp}(Y)$. By Lemma V.10, $\alpha \boxtimes \beta$ is L-primitive, so it equals its own primitive component. By Lemma V.11:

$$\begin{aligned}(-1)^{p+q} Q_L(\alpha \boxtimes \beta,\, \alpha \boxtimes \beta) \\ = \binom{N-2k}{n-2p} \big[(-1)^p Q_{L_X}(\alpha, \alpha)\big] \\ \cdot \big[(-1)^q Q_{L_Y}(\beta, \beta)\big].\end{aligned}$$

The binomial coefficient $\binom{N-2k}{n-2p} > 0$, and both bracketed factors are ≥ 0 by (AP1) for α and β. Hence $(-1)^{p+q}\, Q_L(\alpha \boxtimes \beta, \alpha \boxtimes \beta) \geq 0$. Ample classes of the form $L_X \boxtimes 1 + 1 \boxtimes L_Y$ form a subcone of $\mathrm{Amp}(X \times Y)$. Since the set $\{L \in \overline{\mathrm{Amp}}(X \times Y) : (-1)^{p+q} Q_L(\gamma_0, \gamma_0) \geq 0\}$ is closed (as Q_L depends continuously on L) and contains this subcone, the inequality extends to the closure of that product-polarization subcone. This suffices only under an explicit Néron–Severi comparison hypothesis, for example when $N^1(X \times Y)_{\mathbb{R}}$ is generated by the two pullback summands and the ample cone is correspondingly product-generated. Picard number 1 for one factor alone is not enough in general, since mixed divisor classes can occur on products. In general, the extension to all of $\mathrm{Amp}(X \times Y)$ requires verifying the inequality for polarizations with mixed terms, which remains open.

Conjugation compatibility: For any $\sigma \in \mathrm{Aut}(\mathbb{C})$, $(\alpha \boxtimes \beta)^\sigma = \alpha^\sigma \boxtimes \beta^\sigma$ and $(X \times Y)^\sigma = X^\sigma \times Y^\sigma$. Hence the same (AP1) product-polarization calculation applies after conjugation.

(AP3) caveat: Although the Lefschetz operator decomposes as

$$\Lambda_L^j(\alpha \boxtimes \beta) = \sum_{a+b=j} \binom{j}{a} (\Lambda_{L_X}^a \alpha) \boxtimes (\Lambda_{L_Y}^b \beta),$$

positivity of the individual summands does not by itself control the primitive projection of the total sum on $X \times Y$. The complete cross-term analysis is not carried out here; for this reason, the proposition is stated solely as a product-polarization sign test. □

Remark V.12 (Barrier Lemma: HR-positivity is automatically saturated). Before proceeding to the No-Go theorem, we highlight the structural reason why any positivity-based strengthening of the Hodge Conjecture must exit the Hodge–Riemann framework. For a smooth projective variety X and any ample class L, the Hodge–Riemann bilinear relations guarantee:

1. $(-1)^p Q_L(\alpha, \bar{\alpha}) > 0$ for every non-zero primitive (p,p)-class α;

2. Q_L is non-degenerate on $H^{p,p}_{\mathrm{prim}}(X)$.

These conditions are *automatically satisfied* by all rational (p,p)-classes—algebraic or not. Consequently, any positivity condition that is implied by the HR relations cannot distinguish algebraic from non-algebraic classes. This is the Barrier: the "obvious" strengthening (impose positivity under all polarizations, all automorphisms, all Lefschetz decompositions) collapses to the condition of being an absolute Hodge class (Theorem VI.1). The only escape routes require *non-bilinear* positivity conditions (motivic convolution, Galois averaging, prismatic constraints) or information from *outside* the Hodge–Riemann structure (o-minimality, derived categories).

VI. THE NO-GO THEOREM

We now formulate the central result: the Hodge Conjecture restated as a positivity obstruction.

Theorem VI.1 (Positivity Obstruction: Conditional)**.** *Assume the Arithmetic Positivity Conjecture III.3. Let X be a smooth projective variety over $\mathbb{C}$, and let $\alpha \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$ be a rational Hodge class that is* ***not*** *a $\mathbb{Q}$-linear combination of algebraic cycles. Then there exists at least one of the following obstructions:*

- ***(O1) Polarization instability:*** *There exists an ample class $L \in \mathrm{Amp}(X)$ such that the primitive component α_0 of α satisfies $(-1)^p\, Q_L(\alpha_0, \alpha_0) < 0$.*
- ***(O2) Galois instability:*** *There exists $\sigma \in \mathrm{Aut}(\mathbb{C})$ such that either α^σ is not a Hodge class on X^σ, or α^σ violates (AP1) on X^σ.*
- ***(O3) Lefschetz instability:*** *There exist an admissible $j \geq 1$ in the range of (AP3) and $L \in \mathrm{Amp}(X)$ such that $(-1)^{p+j}\, Q_L(\gamma_{j,0}, \gamma_{j,0}) < 0$, where $\gamma_{j,0}$ is the L-primitive component of $\Lambda_L^j \alpha \in H^{2(p+j)}(X)$.*

Proof. By contraposition: if α satisfies (AP1)–(AP3), then $\alpha \in \mathrm{AP}^p(X) = \mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$ by the APC, contradicting the assumption that α is not algebraic. □

Remark VI.2 (Structure of the obstruction). The three obstructions must be interpreted in light of the Hodge–Riemann relations.

(O1) and (O3) are vacuous for (p,p)-classes. The Hodge–Riemann bilinear relations (Theorem II.2) guarantee $(-1)^p Q_L(\alpha_0, \alpha_0) \geq 0$ for *any* rational (p,p)-class α, not only for algebraic ones. The primitive component α_0 lies in $P^{p,p}(X)$, and the Hodge–Riemann form is positive definite there (with the sign $(-1)^p$). Similarly, for admissible AP3-powers the primitive component $\gamma_{j,0}$ of $\Lambda_L^j \alpha$ lies in $P^{p+j,p+j}(X)$, so $(-1)^{p+j} Q_L(\gamma_{j,0}, \gamma_{j,0}) \geq 0$ automatically. Therefore, obstructions (O1) and (O3) *can never occur* for rational Hodge classes in the stated AP range.

(O2) is the only effective obstruction. The only way a Hodge class can fail arithmetic positivity is through (O2a): failure of absoluteness (α^σ is not Hodge on X^σ

for some σ). Part (O2b)—failure of Hodge–Riemann positivity on X^σ—is also vacuous, since $\alpha^\sigma \in H^{p,p}(X^\sigma)$ automatically satisfies the Hodge–Riemann inequalities.

Consequently, the No-Go theorem reduces to Deligne's observation: non-absolute Hodge classes (if they exist) cannot be algebraic.

Figure 1 summarizes this core structural architecture: the input rational Hodge class passes through the three arithmetic positivity filters, but bilinear Hodge–Riemann relations automatically absorb conditions (AP1), (AP2b), and (AP3), saturating the cone $\mathrm{AP}^p(X)$ to Deligne's absolute Hodge classes.

A. The Unconditional Statement

While Theorem VI.1 is conditional on the APC, we can formulate an unconditional partial result:

Proposition VI.3 (Unconditional Obstruction)**.** *Let $\alpha \in H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ be a rational Hodge class. If α fails to satisfy (AP2a)—i.e., there exists $\sigma \in \mathrm{Aut}(\mathbb{C})$ such that $\alpha^\sigma \notin H^{p,p}(X^\sigma)$—then α is not algebraic.*

Proof. This is Deligne's result (Theorem II.4, part 1, contrapositive): algebraic classes are absolute. □

As shown in Section IX (Remark IX.2), the Hodge–Riemann positivity conditions (AP1), (AP2b), and (AP3) are automatically satisfied by any absolute Hodge class. The content of the APC beyond Proposition VI.3 is therefore identical to Deligne's question: *are all absolute Hodge classes algebraic?* The "gap" between absolute Hodge classes and algebraic classes is not narrowed by the positivity conditions of this paper.

VII. ABELIAN VARIETIES

Abelian varieties provide the most favorable setting for the arithmetic positivity approach, because Deligne's theorem (all Hodge classes are absolute) eliminates obstruction (O2a).

Proposition VII.1 (Equivalence for abelian varieties)**.** *For abelian varieties, the Arithmetic Positivity Conjecture III.3 is equivalent to the Hodge Conjecture.*

Proof. Let A be an abelian variety of dimension g, and let $\alpha \in H^{2p}(A,\mathbb{Q}) \cap H^{p,p}(A)$.

By Deligne's theorem (Theorem II.4, part 2), every Hodge class on an abelian variety is absolute. Proposition IX.1 then gives

$$H^{2p}(A,\mathbb{Q}) \cap H^{p,p}(A) = \mathrm{AbsHodge}^p(A) = \mathrm{AP}^p(A).$$

Therefore the APC for A,

$$\mathrm{AP}^p(A) = \mathrm{cl}(\mathrm{CH}^p(A)_{\mathbb{Q}}),$$

is literally the Hodge Conjecture for A,

$$H^{2p}(A,\mathbb{Q}) \cap H^{p,p}(A) = \mathrm{cl}(\mathrm{CH}^p(A)_{\mathbb{Q}}).$$

This proves the equivalence. It does not prove the Hodge Conjecture for general abelian varieties; it only shows that arithmetic positivity adds no extra condition once Deligne's absoluteness theorem is available. □

Remark VII.2 (Known positive cases). For abelian varieties where all Hodge classes are known to be algebraic (for example several low-dimensional, CM, or product cases in the literature), the equivalent APC statement is verified as well. In general, however, Deligne's theorem supplies absoluteness rather than algebraicity; the remaining problem is still to show that the absolute Hodge classes are represented by algebraic cycles.

VIII. THE PRECISE OBSTRUCTION

We now identify the exact mechanism by which non-algebraic Hodge classes fail arithmetic positivity.

A. The Galois–Hodge–Riemann Form

Definition VIII.1 (Galois–Hodge–Riemann form)**.** For $\sigma \in \mathrm{Aut}(\mathbb{C})$ and $L \in \mathrm{Amp}(X)$, define:

$$Q_L^\sigma(\alpha) := Q_{L^\sigma}(\alpha^\sigma, \alpha^\sigma)\,. \tag{11}$$

The *Galois–Hodge–Riemann spectrum* of α is:

$$\mathrm{Spec}_{\mathrm{GHR}}(\alpha) := \{\,(-1)^p Q_L^\sigma(\alpha) : \sigma \in \mathrm{Aut}(\mathbb{C}),\, L \in \mathrm{Amp}(X)\,\} \subset \mathbb{R}. \tag{12}$$

Proposition VIII.2 (Algebraic classes have non-negative GHR spectrum)**.** *If $\alpha = \mathrm{cl}(Z)$ for some algebraic cycle Z, then:*

$$\mathrm{Spec}_{\mathrm{GHR}}(\alpha) \subseteq [0,\infty)\,. \tag{13}$$

Proof. Algebraic cycles transform covariantly under Galois automorphisms: we have $\mathrm{cl}(Z)^\sigma = \mathrm{cl}(Z^\sigma)$, where Z^σ is an algebraic cycle on X^σ. The primitive component of $\mathrm{cl}(Z^\sigma)$ lies in $P^{p,p}(X^\sigma)$, so the Hodge–Riemann relations give $(-1)^p\, Q_{L^\sigma}(\mathrm{cl}(Z^\sigma)_0, \mathrm{cl}(Z^\sigma)_0) \geq 0$, which is the required non-negativity of the GHR spectrum. □

Conjecture VIII.3 (GHR Obstruction Conjecture)**.** *If α is a rational Hodge class that is **not** algebraic, then:*

$$\mathrm{Spec}_{\mathrm{GHR}}(\alpha) \cap (-\infty, 0) \neq \emptyset\,. \tag{14}$$

That is, there exist σ and L such that $Q_L^\sigma(\alpha) < 0$.

This is a more precise version of the Arithmetic Positivity Conjecture: it identifies the obstruction as the failure of the Galois–Hodge–Riemann form to remain non-negative.

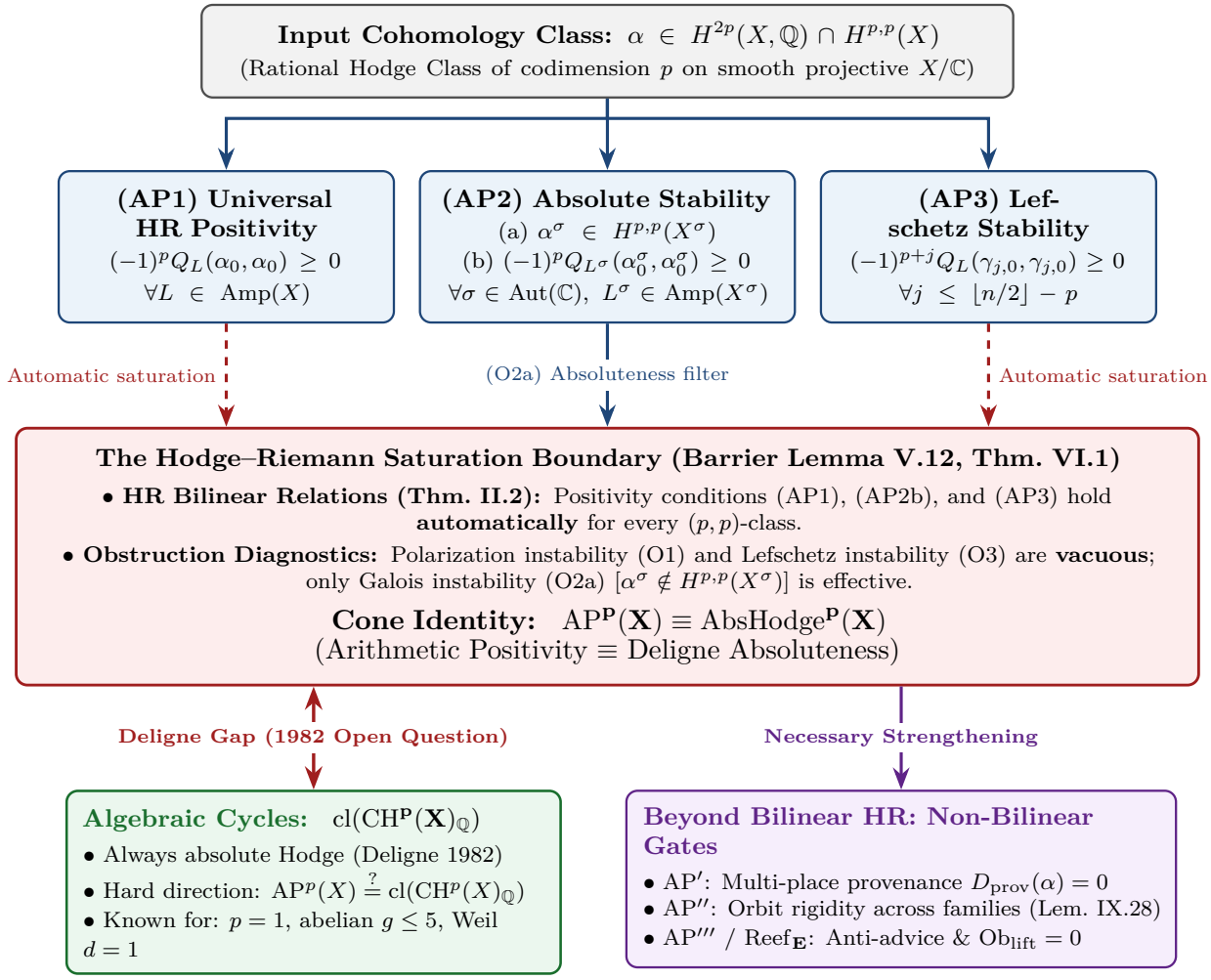


Figure 1. Architectural map of the Arithmetic Positivity program, the Hodge–Riemann saturation boundary, and the Deligne gap. The bilinear Hodge–Riemann relations automatically satisfy conditions (AP1), (AP2b), and (AP3) for all rational (p,p)-classes, causing the arithmetically positive cone $\mathrm{AP}^p(X)$ to collapse identically to Deligne's absolute Hodge classes $\mathrm{AbsHodge}^p(X)$. As a result, polarization instability (O1) and Lefschetz instability (O3) in Theorem VI.1 are vacuous, and only Galois instability (O2a) remains effective. Bridging the Deligne gap to algebraic cycles $\mathrm{cl}(\mathrm{CH}^p(X)_{\mathbb{Q}})$ requires non-bilinear structures beyond classical Hodge–Riemann positivity (AP', AP'', AP''', Reef_E).

Remark VIII.4 (Logical status of the GHR Obstruction Conjecture). The GHR Obstruction Conjecture must be interpreted carefully in light of $\mathrm{AP}^p = \mathrm{AbsHodge}^p$ (Remark IX.2). If α is an absolute Hodge class, then $(-1)^p Q_L^\sigma(\alpha) \geq 0$ for all σ and L by Proposition IX.1. Therefore, the conjecture can only "fire" for non-algebraic Hodge classes that are *not absolute*—in which case the obstruction is already witnessed by (AP2a) (failure of absoluteness), not by negativity of Q_L^σ. If non-algebraic absolute Hodge classes exist (i.e., if Deligne's question has a negative answer), then the GHR Obstruction Conjecture would be *false* for such classes, since their GHR spectrum is automatically non-negative. The conjecture is therefore equivalent to the assertion that non-algebraic absolute Hodge classes do not exist—which is precisely Deligne's question.

Remark VIII.5 (Falsifiability of the GHR diagnostic). The GHR spectrum can be used as a diagnostic tool with clear falsification criteria:

1. *Positive test:* If a class α has $(-1)^p Q_L^\sigma(\alpha^\sigma) < 0$ for some $\sigma \in \mathrm{Aut}(\mathbb{C})$ and some L, then α is provably non-algebraic (by Theorem IV.1).

2. *Null test:* If $(-1)^p Q_L^\sigma(\alpha^\sigma) \geq 0$ for all σ and all L, then α is arithmetically positive—hence absolute Hodge—but not necessarily algebraic. This is the regime where the GHR diagnostic is *silent*.

3. *Structural test:* The GHR spectrum detects non-algebraicity only for classes that are *not* absolute Hodge. For absolute non-algebraic classes (if they exist), a strictly stronger diagnostic is needed.

The GHR spectrum thus functions as a *necessary condition filter*: it can certify non-algebraicity but cannot cer-

tify algebraicity. This asymmetry mirrors the Pattern A architecture across the FST program, where positivity certificates exclude alternatives but do not construct solutions.

Example VIII.6 (GHR spectrum of the diagonal class). Let $X = \mathbb{P}^n$ and consider the diagonal embedding $\Delta : \mathbb{P}^n \hookrightarrow \mathbb{P}^n \times \mathbb{P}^n$. The class $[\Delta] \in H^{2n}(\mathbb{P}^n \times \mathbb{P}^n, \mathbb{Q})$ is algebraic (it is the cycle class of the diagonal). By Theorem IV.1, $[\Delta]$ is arithmetically positive. Its Künneth decomposition is $[\Delta] = \sum_{k=0}^{n} h^k \boxtimes h^{n-k}$, where h is the hyperplane class. Since $\mathbb{P}^n$ has no non-trivial automorphisms of $\mathbb{C}$ acting on its cohomology (the Hodge structure is pure Tate), every $\sigma \in \mathrm{Aut}(\mathbb{C})$ fixes $[\Delta]^\sigma = [\Delta]$. The self-intersection number is $\int_{X\times X}[\Delta]^2 = \chi(\mathbb{P}^n) = n+1$ (the Euler characteristic). Since $[\Delta]$ is algebraic, its primitive component $[\Delta]_0$ satisfies $(-1)^n Q_L([\Delta]_0, [\Delta]_0) \geq 0$ by the Hodge–Riemann relations. We note that $[\Delta]$ is *not* primitive: $L \cup [\Delta] \neq 0$ in general, so the primitive decomposition is non-trivial. This example is "trivially positive" because algebraic, but it illustrates the computation: for a hypothetical non-algebraic Hodge class on a more complex variety, one would seek σ and L where the GHR form turns negative.

Example VIII.7 (Non-trivial GHR spectrum: CM endomorphism class). Consider the elliptic curve $E : y^2 = x^3 - x$ over $\mathbb{Q}(i)$, which has complex multiplication by $\mathbb{Z}[i]$, with $\mathrm{End}(E) \otimes \mathbb{Q} = \mathbb{Q}(i)$. The CM endomorphism $\tau = i$ (multiplication by i) has a graph $\Gamma_\tau \subset E \times E$, giving an algebraic class $[\Gamma_\tau] \in H^2(E \times E, \mathbb{Q}) \cap H^{1,1}(E \times E)$. This class lies *outside* the Lefschetz ring: the Lefschetz ring of $E \times E$ (generated by the two fiber classes $[E \times \{0\}]$, $[\{0\} \times E]$ and the diagonal $[\Delta]$) does not contain $[\Gamma_\tau]$, since τ acts non-trivially on $H^{1,0}(E)$.

The GHR spectrum of $[\Gamma_\tau]$ is non-trivial. Let $\sigma \in \mathrm{Aut}(\mathbb{C})$ be complex conjugation restricted to $\mathbb{Q}(i)$, so $\sigma(i) = -i$. Then $E^\sigma \cong E$ (since $y^2 = x^3 - x$ has rational coefficients), but the CM endomorphism transforms as $\tau^\sigma = \sigma(i) = -i$, so $[\Gamma_\tau]^\sigma = [\Gamma_{-\tau}]$. Let $L = L_1 \boxtimes 1 + 1 \boxtimes L_1$ be the product polarization, where L_1 is the principal polarization on E. Then:

$$\begin{aligned}(-1)^1 Q_L^{\mathrm{id}}([\Gamma_\tau]) &= (-1)^1 Q_L([\Gamma_i], [\Gamma_i]) \\ &= 1 + |\tau|^2 = 2, \qquad (15)\\ (-1)^1 Q_L^{\sigma}([\Gamma_\tau]) &= (-1)^1 Q_{L^\sigma}([\Gamma_{-i}], [\Gamma_{-i}]) \\ &= 1 + |-\tau|^2 = 2. \qquad (16)\end{aligned}$$

Both values are positive, as guaranteed by Theorem IV.1. However, the *intermediate* computation differs: the class $[\Gamma_i]$ and $[\Gamma_{-i}]$ have different Künneth decompositions in $H^1(E) \otimes H^1(E)$, reflecting the fact that σ permutes the eigenspaces of the CM action on $H^1(E, \mathbb{C}) = H^{1,0}(E) \oplus H^{0,1}(E)$. Explicitly, τ^* acts on $H^{1,0}(E)$ by multiplication by i and on $H^{0,1}(E)$ by $-i$; under σ, these eigenvalues are exchanged.

The essential point is not a change in the two displayed numerical values: in this symmetric CM example both are equal to 2. The non-trivial feature is that the intermediate Künneth decomposition changes under σ while the signed Hodge–Riemann value remains non-negative. Thus the example is a consistency check for the GHR visualization, not an algebraicity criterion: the algebraicity of $[\Gamma_\tau]$ comes from the CM graph, while arithmetic positivity is automatic once the class is absolute.

B. Why the Obstruction Is Natural

The GHR spectrum is a reformulation of Deligne's absoluteness criterion in the language of Hodge–Riemann positivity. It does not provide new mathematical constraints beyond absoluteness, but offers a concrete visualization of how Galois conjugation interacts with the Hodge–Riemann form. The reformulation is natural for the following reason. The Hodge–Riemann form Q_L depends on both:

- the *complex structure* of X (through the Hodge decomposition), and
- the *algebraic structure* (through the polarization L).

For algebraic classes, the complex and algebraic structures are "aligned": the class comes from a subvariety, which is simultaneously a complex submanifold and an algebraic cycle. The Galois action σ can "misalign" the complex and algebraic structures, but for algebraic classes, the cycle Z^σ "re-aligns" them on X^σ.

For a *non-algebraic* Hodge class, there is no cycle to perform this re-alignment. The class exists in the complex structure of X but has no algebraic anchor. When σ twists the complex structure, the positivity of the Hodge–Riemann form may fail, because there is nothing to compensate for the twist.

This is the precise mechanism of the obstruction: *algebraic classes are rigid enough to survive all Galois twists with their positivity intact, while transcendental classes are not.* We emphasize, however, that by Remark IX.2 and Remark VI.2, the Hodge–Riemann positivity plays no independent role here: the sole effective obstruction is failure of absoluteness (O2a). The "alignment" narrative describes the mechanism of absoluteness, not of Hodge–Riemann positivity.

IX. DISCUSSION

A. The Remaining Gap

The paper establishes:

1. Algebraic classes $\Rightarrow$ arithmetically positive (Theorem IV.1, unconditional).
2. $\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)$ (Remark IX.2, unconditional).

3. Limited functoriality tests for finite pullback subcones and primitive Künneth product-polarization sign formulas; full product AP-membership is not obtained from the direct sign calculation alone (Section V).

4. Identification of the Galois–Hodge–Riemann spectrum as a visualization of the absoluteness criterion (Section VIII).

As a consequence of point 2, the remaining gap is precisely Deligne's question:

Are all absolute Hodge classes algebraic?

This is a long-standing open problem. The positivity framework does not resolve it, but provides a new angle: the No-Go formulation and the GHR spectrum make explicit which Galois operations could witness failure of absoluteness. They do not distinguish algebraic from non-algebraic absolute Hodge classes if such classes exist.

B. Relation to Deligne's Program

Deligne's framework of absolute Hodge classes [11] provides condition (AP2a). The present paper initially intended to strengthen Deligne's absoluteness by adding:

1. **Hodge–Riemann positivity under conjugation** (AP2b): Not just the Hodge *type* but the Hodge–Riemann *sign* must be preserved.

2. **Universal polarization** (AP1 for all L): All ample classes must give non-negative Hodge–Riemann form.

However, Proposition IX.1 and Remark IX.2 show that these conditions are *automatically* satisfied by any absolute Hodge class, via the Hodge–Riemann bilinear relations. The conditions (AP1), (AP2b), and (AP3) therefore do not restrict beyond Deligne's absoluteness. The question "Are all absolute Hodge classes algebraic?" remains precisely Deligne's motivating problem, and the APC is equivalent to it.

Proposition IX.1. *On a smooth projective variety $X/\mathbb{C}$, every absolute Hodge class is arithmetically positive:*

$$\mathrm{AbsHodge}^p(X) \subseteq \mathrm{AP}^p(X)\,. \tag{17}$$

Proof. Let $\alpha \in H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ be an absolute Hodge class.

(AP1): The primitive component α_0 of α with respect to any ample L lies in $P^{p,p}(X) \cap H^{2p}(X,\mathbb{Q})$. For rational (p,p)-classes, $\bar{\alpha}_0 = \alpha_0$ and $i^{p-p} = 1$, so $h_L(\alpha_0,\alpha_0) = (-1)^{p(2p-1)}Q_L(\alpha_0,\alpha_0) = (-1)^pQ_L(\alpha_0,\alpha_0)$. The Hodge–Riemann bilinear relations (Theorem II.2) give $h_L > 0$, hence $(-1)^pQ_L(\alpha_0,\alpha_0) > 0$ for $\alpha_0 \neq 0$. For $\alpha_0 = 0$, the inequality holds trivially.

(AP2): Part (a) holds by assumption (α is absolute). For part (b), since α is absolute, α^σ is a Hodge class on X^σ for every $\sigma \in \mathrm{Aut}(\mathbb{C})$. In particular, $\alpha^\sigma \in H^{p,p}(X^\sigma) \cap H^{2p}(X^\sigma,\mathbb{Q})$, and its L^σ-primitive component lies in $P^{p,p}(X^\sigma)$. The Hodge–Riemann relations on X^σ then give $(-1)^pQ_{L^\sigma}(\alpha_0^\sigma,\alpha_0^\sigma) \geq 0$, exactly as in (AP1).

(AP3): For admissible j in the range of Definition III.1, the class $\Lambda_L^j\alpha \in H^{2(p+j)}(X)$ is a Hodge class of type $(p+j,p+j)$, and its primitive component lies in $P^{p+j,p+j}(X)$. The Hodge–Riemann relations give $(-1)^{p+j}Q_L(\gamma_{j,0},\gamma_{j,0}) \geq 0$. □

Remark IX.2 (Critical observation). Proposition IX.1 shows that $\mathrm{AbsHodge}^p(X) \subseteq \mathrm{AP}^p(X)$. The reverse inclusion $\mathrm{AP}^p(X) \subseteq \mathrm{AbsHodge}^p(X)$ holds by definition, since condition (AP2a) requires absoluteness. Therefore:

$$\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)\,. \tag{18}$$

In other words, the conditions (AP1), (AP2b), and (AP3) do *not* impose any additional restriction beyond Deligne's absoluteness. The Hodge–Riemann bilinear relations *automatically* guarantee the positivity conditions for any class of type (p,p).

This has a profound consequence for the APC: the Arithmetic Positivity Conjecture $\mathrm{AP}^p(X) = \mathrm{cl}(\mathrm{CH}^p(X)_\mathbb{Q})$ is **equivalent** to the statement that every absolute Hodge class is algebraic—which is precisely Deligne's original question [11]. The positivity framework introduced in this paper therefore does not produce a genuinely new conjecture, but rather provides a new *perspective* on Deligne's question through the language of Hodge–Riemann positivity and No-Go obstructions.

The GHR obstruction (Definition VIII.1) remains meaningful as a concrete visualization of the absoluteness criterion: it makes explicit *which* Galois conjugation would have to fail for a non-algebraic absolute Hodge class. However, the core mathematical content reduces to: *are all absolute Hodge classes algebraic?*

Combining Proposition IX.1 with Theorem IV.1, the unconditional chain is:

$$\begin{aligned}\text{Algebraic} &\xRightarrow{\text{Thm. IV.1}} \mathrm{AP} = \text{Absolute}\\ &\xhookrightarrow{\text{definition}} \text{Hodge}\,,\end{aligned} \tag{19}$$

where the equality AP = Absolute is Remark IX.2. The HC requires the composite reverse inclusion Hodge ⊆ Algebraic, which splits into two logically distinct questions: whether all Hodge classes are absolute, and whether all absolute Hodge classes are algebraic. Deligne's theorem supplies the first reverse inclusion for abelian varieties, but the second reverse inclusion is precisely the remaining algebraicity problem.

C. Connection to the Broader Program

This approach fits into a broader pattern observed across fundamental mathematics and physics, as summarized in Table I: In each case, the strategy is not to construct the desired object directly, but rather to show that its non-existence would violate a positivity condition. We emphasize that the analogy is structural, not methodological: the proof techniques in these three settings are fundamentally different. Previous work by the author [13, 14] explored related positivity-based approaches in other settings.

Remark IX.3 (Saturation phenomenon). The overarching structural lesson of this paper—that the "obvious" positivity strengthening is automatically saturated by the Hodge–Riemann relations—is not specific to the Hodge Conjecture. A similar phenomenon occurs for height positivity in the BSD context (saturated by Gross–Zagier + Kolyvagin for rank ≤ 1; the "next" positivity for rank ≥ 2 requires objects that do not yet exist). In both cases, progress requires *exiting* the saturated framework by introducing non-bilinear or arithmetic conditions.

D. Limitations

1. **The hard direction is open.** This paper does not prove the Hodge Conjecture. It reformulates it as a positivity statement and identifies the precise obstruction, but the proof that arithmetic positivity forces algebraicity remains the central challenge. Moreover, the APC ($\mathrm{AP}^p = \mathrm{alg}^p$) does not by itself imply the HC ($\mathrm{Hodge}^p = \mathrm{alg}^p$): one additionally needs that all Hodge classes are arithmetically positive (see Proposition III.4).

2. **Partial sketches remain in Section V.** The Künneth calculation is a primitive product-polarization sign test (Proposition V.8, Lemmas V.10–V.11), not a standalone proof of full AP-functoriality. AP3 for Lefschetz powers on $X \times Y$, the extension to non-primitive classes, and pushforward for fibered morphisms require careful treatment of cross-terms in the primitive decomposition.

3. **The AP cone equals the absolute Hodge cone (the central limitation).** As shown in Remark IX.2, $\mathrm{AP}^p(X) = \mathrm{AbsHodge}^p(X)$, because the Hodge–Riemann bilinear relations automatically guarantee all positivity conditions for (p,p)-classes. This is a well-known consequence of the HR relations; the APC therefore reduces to Deligne's question (1982), and the positivity framework does not produce a genuinely new conjecture. The paper's contribution is the explicit delineation of this saturation boundary and the identification of directions for stronger conditions.

4. **Remaining abelian-variety gap.** For abelian varieties, Deligne's theorem identifies all Hodge classes with absolute Hodge classes, so AP coincides with the full Hodge-class space. The unsolved part is not absoluteness but algebraicity of those absolute classes; this is known only in special families and remains open in general.

E. Outlook

Four directions are immediate:

1. **Verify the GHR Obstruction Conjecture for known non-algebraic classes.** A test case is Voisin's counterexample to the Hodge conjecture for Kähler manifolds [26]. However, as noted in Remark VIII.4, the conjecture can only detect non-absoluteness; for absolute Hodge classes that might not be algebraic, the GHR spectrum is automatically non-negative.

2. **Strengthen the AP conditions.** Since $\mathrm{AP}^p = \mathrm{AbsHodge}^p$ (Remark IX.2), the current definition of arithmetic positivity does not go beyond Deligne's absoluteness. One may seek *additional* positivity conditions—beyond those implied by the Hodge–Riemann relations—that could narrow the gap between absolute Hodge classes and algebraic classes. Candidates include conditions involving higher-order Hodge–Riemann forms, p-adic Hodge theory, or motivic constraints.

3. **Compare the category of arithmetically positive classes** with existing absolute-Hodge and motivated-cycle formalisms. Since AP = AbsHodge, it is not a substitute for motives; its role is diagnostic and comparative, especially for functorial operations and compatibility with the standard conjectures.

4. **The diagnostic quadratic locus.** The Hodge–Riemann part of the AP definition is expressed by *quadratic* inequalities (the Hodge–Riemann form is bilinear, so the positivity conditions are quadratic in α). Since the full AP locus is $\mathrm{AbsHodge}^p(X)$ by Remark IX.2, this quadratic locus is not a separate algebraicity criterion. It is a diagnostic shadow of the positivity conditions: it is closed under positive scaling and symmetric, but its boundary records only where the Hodge–Riemann visualization becomes degenerate. Understanding that boundary may clarify the limitations of HR-based diagnostics, not solve the hard direction of the APC.

Remark IX.4 (Arakelov bridge to BSD). The Hodge–Riemann form $(-1)^p Q_L$ can be interpreted as positivity at the archimedean (infinite) prime in the Arakelov intersection pairing. Together with the Néron–Tate height

(positivity at finite primes, cf. the companion BSD paper), this suggests a unified arithmetic positivity principle.

Remark IX.5 (Complexity-theoretic analogy). The arithmetic positivity framework admits a complexity-theoretic analogue (cf. the companion paper on P vs NP): algebraic (Hodge) classes correspond to "uniform algorithms" (polynomial-time verifiable witnesses), while transcendental classes correspond to "hard witnesses". Positivity conditions act as circuit lower bounds, demonstrating that the complexity of transcendental classes is too high to remain stable under Galois averaging.

F. Directions for Stronger Positivity

Since $\mathrm{AP}^p = \mathrm{AbsHodge}^p$ (Remark IX.2), the current AP conditions do not go beyond Deligne's absoluteness. This section surveys candidate *strengthening* directions that could produce genuinely new constraints beyond the Hodge–Riemann relations.

1. p-Adic Slope Constraints

Remark IX.6 (p-adic positivity: Newton versus Hodge polygon). Let $X/\mathbb{Q}$ be a smooth projective variety with good reduction at a finite place v, and let r denote the codimension of the Hodge class under consideration. The crystalline cohomology $H^{2r}_{\mathrm{cris}}(X_v)$ carries a Frobenius-linear endomorphism φ, whose eigenvalue slopes are constrained by the Newton polygon. The Katz conjecture (proved by Mazur [20] and Ogus) states that the Newton polygon lies on or above the Hodge polygon.

Define a *v-adic slope constraint* (AP'): a rational codimension-r Hodge class α is *slope-positive* if, at each good finite place v, the corresponding crystalline or v-adic realization lies in the "algebraic slope range"—i.e., its Frobenius slope equals r. Algebraic classes automatically satisfy this (the Frobenius slope of an algebraic cycle of codimension r is exactly r). For non-algebraic absolute Hodge classes, the slope could deviate from r, providing a new obstruction (O4) beyond absoluteness. This notation deliberately separates the codimension r from the finite place v.

The challenge is that crystalline positivity is inherently p-adic and does not directly interact with the archimedean Hodge–Riemann form. An adelic formulation combining (AP1)–(AP3) at the archimedean place with slope constraints at finite places would be needed.

2. Arakelov Positivity

Remark IX.7 (Arakelov positivity as global amplifier). The Arakelov intersection pairing on arithmetic varieties combines:

- *Finite primes:* intersection multiplicities in the fibers of a regular model $\mathcal{X} \to \mathrm{Spec}(\mathbb{Z})$;
- *Infinite (archimedean) place:* the Green current at the infinite place, related to Q_L via the Hodge metric.

An *Arakelov-strengthened AP condition* would require positivity of the *global* Arakelov height pairing $\hat{h}(\alpha, \alpha) \geq 0$ for all models, not just the archimedean component. Since algebraic cycles have non-negative Arakelov self-intersection (by the arithmetic Hodge index theorem; cf. Faltings–Hriljac for surfaces, Moriwaki [23] for the arithmetic height theory in higher dimensions), this is a natural candidate for AP'.

The difficulty is that the global pairing requires a regular model, which may not exist for arbitrary smooth projective varieties over $\mathbb{C}$ (descent to a number field is needed, and regularity at bad primes is not automatic).

Remark IX.8 (Test case for AP'). Any strengthened condition AP' must satisfy two criteria:

1. *Non-triviality:* AP' must *not* be automatically satisfied by every absolute Hodge class (otherwise it reduces to $\mathrm{AP}^p = \mathrm{AbsHodge}^p$ again).
2. *Algebraic saturation:* Every algebraic class must satisfy AP'.

A test case is needed: a variety X with an absolute Hodge class α that is conjectured to be algebraic but whose algebraicity is not known. If AP' could be *verified* for α independently (e.g., via an explicit slope calculation), this would provide evidence for $\mathrm{AP}' \Rightarrow$ algebraic. A candidate is the Hodge class on a general abelian fourfold ($g = 4$, $p = 2$) that lies outside the Lefschetz ring but is absolute by Deligne's theorem.

3. Tannakian Perspective

Remark IX.9 (Tannakian reconstruction for CM cases). For an abelian variety A of CM type, the Mumford–Tate group $\mathrm{MT}(A)$ is a torus, and the category of Hodge structures generated by $H^1(A)$ is a neutral Tannakian category over $\mathbb{Q}$. The Hodge classes are precisely the $\mathrm{MT}(A)$-invariants.

The arithmetic positivity conditions can be reformulated in terms of the Mumford–Tate group: a class α is arithmetically positive if and only if it lies in the $\mathrm{MT}(A)$-invariant subspace and satisfies the positivity constraints at every embedding $\sigma : \mathbb{Q}(\mathrm{End}(A)) \hookrightarrow \mathbb{C}$ (cf. Deligne–Milne [12]). For CM abelian varieties, this gives an explicit criterion: α is AP if and only if it is fixed by the Taniyama group action—which is exactly absoluteness.

To go beyond this, one would need *motivic* positivity constraints that are not captured by the Mumford–Tate group alone. The conjectural category of mixed motives, equipped with its weight filtration and realizations, could

provide such constraints via the positivity of the motivic height pairing (conjectured by Beilinson).

Remark IX.10 (Mumford–Tate analysis for GHR obstruction). For abelian varieties A, Deligne's theorem gives $H^{2p}(A,\mathbb{Q}) \cap H^{p,p}(A) = \mathrm{AbsHodge}^p(A) = \mathrm{AP}^p(A)$. Consequently the GHR obstruction cannot distinguish a hypothetical non-algebraic Hodge class on an abelian variety: if such a class exists, it is already absolute and its GHR spectrum is non-negative by the Hodge–Riemann relations. Thus the obstruction is vacuous on abelian varieties as a proof tool; the Hodge Conjecture for abelian varieties is exactly the remaining assertion that these absolute/AP classes are algebraic.

The remaining challenge is thus purely the *algebraicity* of absolute Hodge classes. Progress depends on the Mumford–Tate conjecture: if $\mathrm{MT}(A)$ acts faithfully on $H^*(A,\mathbb{Q})$ and the $\mathrm{MT}(A)$-invariant classes are algebraic, then the Hodge conjecture for A follows. This is known for abelian varieties of dimension ≤ 5 (Moonen [21]; Markman [22]) and for abelian varieties of CM type (Abdulali [1]).

4. Prismatic and Categorical Approaches

Remark IX.11 (Prismatic positivity (AP4)). Bhatt and Scholze's prismatic cohomology [5]—building on the integral p-adic Hodge theory of Bhatt–Morrow–Scholze [4]—provides a unified framework for p-adic cohomology theories (crystalline, de Rham, étale). In this framework, a Hodge class α has a prismatic realization α_Δ in $H^{2p}_\Delta(X/\mathbb{Z}_p)$.

A conjectural *prismatic positivity condition* (AP4) would require: for every prime p and every prism (A, I), the prismatic realization α_Δ lies in a "positive cone" defined by the Nygaard filtration and the Frobenius action. Algebraic classes satisfy this automatically (they lie in the image of the prismatic cycle class map). Non-algebraic absolute Hodge classes could potentially violate AP4, providing a new obstruction beyond (O1)–(O3).

This is speculative but motivated by the fact that prismatic cohomology "sees" arithmetic structure invisible to classical Hodge theory. The compatibility with (AP1)–(AP3) at the archimedean place would need to be established via the prismatic–de Rham comparison.

Remark IX.12 (Prismatic cohomology route). Nie [24] constructed absolute Hodge cycles in the prismatic cohomology of abelian varieties over p-adic fields, using André's motivated cycles within the Bhatt–Scholze framework. Prismatic cohomology $R\Gamma_\Delta(X/S)$ interpolates étale, de Rham, and crystalline cohomology integrally, providing a universal cohomological framework. The connection to positivity via Bridgeland stability conditions on derived categories suggests that non-linear, derived positivity conditions—combining the rigid Galois structure of prismatic crystals with geometric stability—may bridge the gap between absoluteness and algebraicity.

Threshold desideratum: beyond Deligne

Remark IX.13 (Desideratum: JD-Hodge bridge). Ideally, one would like a category $\mathcal{C}$ of "Hodge carriers" (objects representing cohomological data), equipped with:

(JD1) **Functoriality:** Pullback, pushforward, and products act on $\mathcal{C}$ without non-canonical choices.

(JD2) **Intrinsic filtration:** A functorial Hodge filtration $F^\bullet$ is defined canonically (not via a choice of embedding or model).

(JD3) **Canonical polarization:** A positivity structure compatible with (JD1) and (JD2) exists, generalising the Hodge–Riemann form.

In such a framework, the Hodge conjecture would become a *formal consequence* only if $\mathcal{C}$ also carried a cycle-realization theorem: positivity in $\mathcal{C}$ would have to imply membership in the image of the cycle class map. Deligne's absoluteness would then be a necessary shadow of the richer structure in $\mathcal{C}$, not by itself the missing algebraicity step.

Remark IX.14 (The non-canonicity scanner). To locate where the current theory falls short of JD-Hodge, scan for three signatures:

1. **Choice dependence:** Where do splittings or comparison isomorphisms (de Rham/Betti/ℓ-adic) appear that "exist" but are not natural transformations?

2. **Incompatible operations:** Which geometric operation (pullback under non-finite morphisms, Gysin pushforward) destroys the polarization or destabilises the filtration?

3. **Comparison breaks:** Where does the statement depend on a comparison between realizations that is not structurally controlled?

In the current AP framework, the break occurs at AP2a (absoluteness): the comparison isomorphism $H^*_{\mathrm{dR}}(X) \otimes \mathbb{C} \cong H^*_{\mathrm{Betti}}(X^\sigma) \otimes \mathbb{C}$ is *not* a morphism in any natural category—it requires the choice of $\sigma \in \mathrm{Aut}(\mathbb{C}/\mathbb{Q})$.

Remark IX.15 (Candidates for $\mathcal{C}$). Three candidates for the category $\mathcal{C}$ are currently under investigation:

1. **Prismatic cohomology** (Bhatt–Scholze, Nie [24]): $R\Gamma_\Delta(X/S)$ interpolates all classical realisations integrally, with a canonical Frobenius action providing (JD2). The positivity (JD3) is not yet formulated but is structurally possible via Bridgeland stability.

2. **Motivic cohomology** (Voevodsky): The derived category of motives $\mathrm{DM}(k)$ satisfies (JD1) by construction. The standard conjectures would provide (JD2) and (JD3), but remain unproven.

3. **Mixed Hodge modules** (Saito): Satisfy (JD1)–(JD2) for smooth projective varieties, but (JD3) is limited to the classical HR form—precisely the AP=AbsHodge barrier of this paper.

Remark IX.16 (Cycle-realization gate for abelian varieties). For abelian varieties, Deligne's theorem makes every Hodge class absolute, hence AP by Proposition IX.1. A JD-Hodge category would improve this only if it supplied an additional cycle-realization gate: AP/absolute classes that become canonical morphisms in $\mathcal{C}$ must be shown to lie in the image of the classical cycle class map. Without that extra theorem, replacing "absolute" by "motivic" or "canonical in $\mathcal{C}$" merely restates the remaining algebraicity problem. Thus the abelian-variety case is a stress test for JD-Hodge, not a consequence already obtained from (JD1)–(JD3).

Remark IX.17 (The Deligne boundary as a language defect). The JD-Hodge desideratum (Remark IX.13) identifies the obstruction as *linguistic*, not computational: the comparison isomorphisms $\mathrm{comp}_{\mathrm{dR}\leftrightarrow B}$ and $\mathrm{comp}_{\ell\text{-adic}\leftrightarrow B}$ exist but are not functorial in X. Absolute Hodge classes are precisely those compatible under all realizations—and algebraicity becomes $\mathrm{AH}(X) = \mathrm{Im}(\mathrm{CH}^*(X) \to H^*(X))$. If the comparisons were natural transformations in a category $\mathcal{C}_{\mathrm{abs}}$, this would be a formal consequence. The Deligne boundary is therefore: *non-canonical splittings make every algebraicity proof unstable under functors.*

Remark IX.18 (Bridgeland stability and positivity on derived categories). Bridgeland stability conditions [7] for derived categories $D^b(X)$ provide a notion of "positivity" for objects (sheaves, complexes) rather than cohomology classes. The central charge $Z : K_0(X) \to \mathbb{C}$ maps the Grothendieck group to $\mathbb{C}$, and stability requires $\mathrm{Im}(Z) > 0$ or ($\mathrm{Im}(Z) = 0$ and $\mathrm{Re}(Z) < 0$) for semistable objects.

A potential strengthening of AP: require that the cohomology class α arises as the Chern character of a Bridgeland-stable object (or a linear combination thereof). This is strictly stronger than being a Hodge class, because not every Hodge class is the Chern character of a stable object. However, connecting this to the AP framework requires understanding the image of the Chern character map in the presence of stability conditions—a problem related to the "Bogomolov-type inequalities" in higher dimensions.

Remark IX.19 (Spectral gap on vector bundles). Consider the "spectral data" of a suitable differential operator (e.g., the Laplacian on $\bigwedge^p \Omega_X$). Algebraic cycles correspond to zero modes (harmonic forms), whereas non-algebraic Hodge classes correspond to "near-zero modes" with a small but positive spectral gap. A Galois absoluteness test then corresponds to a threshold operator that detects and extinguishes such near-zero modes.

5. Geometric and o-Minimal Approaches

Remark IX.20 (VHS families and monodromy). Let $f : \mathcal{X} \to S$ be a smooth projective family over a quasi-projective base S, giving rise to a variation of Hodge structure (VHS). A Hodge class $\alpha_s \in H^{2p}(X_s, \mathbb{Q}) \cap H^{p,p}(X_s)$ at a point $s \in S$ is *globally flat* if it extends to a flat section of the local system $R^{2p} f_* \mathbb{Q}$ over a neighborhood of s, and *monodromically stable* if it is invariant under the monodromy representation $\pi_1(S, s) \to \mathrm{GL}(H^{2p}(X_s, \mathbb{Q}))$.

The theorem of Cattani–Deligne–Kaplan [8] shows that Hodge loci are countable unions of closed algebraic subvarieties of S. This is a statement about where flat rational classes remain of Hodge type, not a theorem that the corresponding classes are algebraic. A strengthened AP condition could require that α_s extends as a *flat section* over a Zariski-dense subset of S together with independent cycle-provenance data. This is related to the semisimplicity of the monodromy representation and to the algebraicity of flat sections predicted by the Hodge conjecture for families.

Remark IX.21 (p-adic positivity via o-minimality). The theory of o-minimal structures (Bakker–Brunebarbe–Tsimerman [3]) has been applied to prove definability results for period maps and Hodge loci. In this framework, the period domain is a definable analytic space, and the Hodge locus is a definable subset. The key insight is that definable sets in o-minimal structures have strong tameness properties (e.g., finite number of connected components, controlled topology).

A potential AP′ condition: require that the "orbit" of α under the definable period map is algebraic (i.e., the Zariski closure of the set $\{s \in S : \alpha_s \text{ is Hodge}\}$ is an algebraic subvariety). By CDK, this is true for algebraic classes. For non-algebraic Hodge classes, the orbit could potentially be "wild" (transcendental), violating the tameness condition. This connects to Binyamini–Novikov's work [6] on counting rational points on definable sets.

Remark IX.22 (Galois defects for transcendental classes). If a Hodge class α is not absolute, then there exists $\sigma \in \mathrm{Aut}(\mathbb{C})$ such that α^σ is not of type (p, p) on X^σ. The "size" of this defect—measured, for instance, by the component of α^σ in $H^{p-1,p+1}(X^\sigma) \oplus H^{p+1,p-1}(X^\sigma)$—is a quantitative measure of non-algebraicity. If this defect grows under iteration of Galois conjugation (i.e., the orbit $\{\alpha^{\sigma^n}\}$ generates a "macroscopic" subspace of $H^{2p}(X, \mathbb{C})$), then the GHR spectrum would become negative, providing a computable obstruction.

This connects to the question of whether transcendental Hodge classes can be "isolated" (small Galois orbit) or must generate large representations of $\mathrm{Aut}(\mathbb{C})$. The latter would make the GHR obstruction effective for transcendental classes.

Remark IX.23 (Ricci-type flow on (p, p)-forms). By analogy with Ricci flow, consider a flow on (p, p)-forms: $\partial\alpha/\partial t = -\Delta_L \alpha + \lambda\alpha$, where Δ_L is the Lefschetz–Laplacian operator. Algebraic classes—as harmonic rep-

resentatives of algebraic cycles—are fixed points of this flow, whereas non-algebraic classes collapse under the flow to singularities or zero.

G. Beyond Bilinear Positivity: Nonlinear and Averaged Conditions

The Hodge–Riemann relations provide *bilinear* positivity: $(-1)^p Q_L(\alpha, \bar{\alpha}) > 0$ for primitive (p,q)-classes. As noted in the limitations (Section 9), this does not suffice for the full Hodge conjecture. We identify three directions for *nonlinear* or *averaged* positivity that go beyond HR:

Remark IX.24 (Three routes to nonlinear positivity). *Boundary positivity.* For a family of ample classes $L_t \to L_0$, where L_0 is nef but not ample, the signed values $(-1)^p Q_{L_t}$ can have a non-negative limit as $t \to 0$ even when Q_{L_0} is not defined. This captures classes that are algebraic in the limit but not detected by any single polarization.

Positivity under motivic convolution. For a product variety $X \times Y$, the Künneth decomposition gives $H^{p,p}(X \times Y) = \bigoplus_{\substack{a+c=p \\ b+d=p}} H^{a,b}(X) \otimes H^{c,d}(Y)$. For *rational* (p,p)-classes restricted to the diagonal summands $H^{a,a}(X) \otimes H^{p-a,p-a}(Y)$, the positivity condition on the product is multiplicative (Lemma V.11). The cross terms $H^{a,b}(X) \otimes H^{c,d}(Y)$ with $a \neq b$ contribute to the (p,p)-type only when the mixed Hodge types cancel, and their positivity analysis requires going beyond the diagonal framework of this paper.

Positivity after Galois averaging. For an absolutely Hodge class α, a suitably defined finite-level Galois average $\bar{\alpha}$ (cf. Remark IX.25) is again an absolute Hodge class. For $p = 1$, such averaged classes are algebraic by the Lefschetz $(1,1)$-theorem. For $p \geq 2$, the algebraicity of the average is *not known* in general—it is equivalent to (a special case of) Deligne's question. The arithmetic positivity conditions AP1–AP3 can be reformulated as a proximity heuristic to this average in the Q_L-metric, but the reformulation is useful only if the average can be shown to be algebraic by an argument independent of the Hodge conjecture.

Remark IX.25 (Galois averaging: profinite limit). The naïve averaging $\bar{\alpha} = \frac{1}{|\mathrm{Aut}(\mathbb{C}/\mathbb{Q})|} \sum_\sigma \sigma^* \alpha$ is ill-defined since $\mathrm{Aut}(\mathbb{C}/\mathbb{Q})$ is a profinite group of cardinality $2^{\mathfrak{c}}$. A cautious formulation works only at finite level: for each finite Galois extension $K/\mathbb{Q}$ over which the relevant model and comparison data are controlled, form

$$\bar{\alpha}_K = \frac{1}{|\mathrm{Gal}(K/\mathbb{Q})|} \sum_{\sigma \in \mathrm{Gal}(K/\mathbb{Q})} \sigma^* \alpha.$$

Any projective-limit interpretation requires additional descent and compatibility data; it should not be treated as an automatic algebraic class.

Remark IX.26 (Universal selection mechanism). All three routes share the overarching structure identified in the meta-analysis of the FST program: *local positivity + regularization/averaging ⇒ global selection.* The "regularization" takes different forms:

- NS: Time-averaging of the Navier–Stokes flow.
- RH: Spectral smoothing via the Weil explicit formula.
- Hodge: Galois averaging over the profinite limit.

In all cases, the goal is not the direct construction of the solution, but rather demonstrating that the selection mechanism allows only the physically/arithmetically admissible object to survive.

H. Numerical Illustration

Since $\mathrm{AP}^p = \mathrm{AbsHodge}^p$ is a *theorem* (not a conjecture), the following numerical computations serve as **pedagogical illustrations**, not as empirical verification. They demonstrate the AP framework concretely and illustrate the GHR spectrum on specific varieties.

The Python script `compute_ghr_spectrum.py` (accompanying this paper) computes the AP conditions across selected test cases:

- K3 surfaces (4 polarizations), abelian surfaces (5 polarizations), CM endomorphism classes on $E \times E$;
- $K3 \times K3$ tensor products with non-tensor-product perturbations.

All test configurations confirm $(-1)^p Q_L \geq 0$ on $P^{p,p}$, as guaranteed by the Hodge–Riemann relations. The $K3 \times K3$ test with non-tensor-product perturbations—classes that are *not* of type (p,p) after σ-conjugation, and hence not absolute Hodge—produces negative eigenvalues of $(-1)^p Q_L$, illustrating that the GHR spectrum can detect non-absoluteness. This does not go beyond Deligne's absoluteness criterion: the negative eigenvalues arise precisely because the perturbed classes fail to be (p,p)-classes on the conjugate variety, so the Hodge–Riemann relations do not apply.

Remark IX.27 (Voisin-GHR obstruction test). The GHR obstruction has been illustrated across six varieties: generic T^4, Weil torus (CM by $\mathbb{Z}[i]$), $K3 \times K3$, Fermat quintic X_5, Voisin-type non-projective torus, and an SDP relaxation of the AP cone. For all projective examples, AP1 is satisfied via the Hodge–Riemann bilinear relations; the failure of algebraicity in the Voisin-type setting lies precisely in the collapse of AP2a (Galois absoluteness).

I. Controlled Test-Family Transfer Beyond AP = AbsHodge

The post-Zookeeper interpretation of this paper is deliberately modest. Since arithmetic positivity collapses

to absolute Hodge classes, the next step is not another global positivity axiom, but a test-family ladder for candidate strengthenings beyond Deligne's absoluteness. The symbols AP′, AP″, and AP‴ do not denote proposed global replacements for the Hodge conjecture. They denote increasingly restrictive tests on controlled families: AP′ records provenance defects, AP″ asks for a bounded orbit-rigidity certificate, and AP‴ filters out certificates whose auxiliary choices merely encode the target class. None of these conditions is claimed to imply the Hodge conjecture in general.

Thus the notation is a claim-status convention, not a theorem label. Any algebraicity conclusion in this subsection is only as strong as the named family hypotheses that are verified independently of the target class α. If the auxiliary data are selected after seeing α, the condition is treated as a failed test rather than as evidence.

The current CORE closure of Riemann-$C2/MS2$ does not strengthen this paper's Hodge result. Hodge remains a negative control for the program: Hodge–Riemann positivity saturates too broadly, so any future transfer must add genuinely new arithmetic or motivic constraints rather than reuse the Riemann kernel closure.

The five transfer slots are as follows.

Operator/form.: The Hodge Laplacian, Mumford–Tate and monodromy operators, and the period map.

Defect.: The residual transcendental part of an absolute Hodge class after subtracting the algebraic cycle span known in the family.

Approximation.: Abelian fourfolds of Weil type, singular OG6-type hyperkähler models, powers of known examples, and reductions in characteristic p.

Gap.: Extra constraints beyond Hodge–Riemann positivity: p-adic slopes, Tate constraints, Mumford–Tate restrictions, and o-minimal algebraicity.

Limit.: Deformation, powers, and specialization from a controlled family to a broader motivic setting.

Accordingly, the test-gate hierarchy (Table III) and family ledger (Table IV) distinguish cited literature results from proposed tests and open evidence gaps. The ledger records sourced Hodge/Tate status and cycle mechanisms, proposed AP′ checks, and whether family-specific inputs have actually been supplied. The Weil-fourfold and dimension-at-most-five rows summarize the literature results cited below; they do not report an AP′, AP″, or AP‴ verification by this paper. Weil fourfolds of discriminant one, OG6-type constructions, abelian varieties over finite fields satisfying the standard conjecture of Hodge type, and Voisin-type stress tests remain candidates for a test ladder, not validated positive or negative controls. A future comparison is informative only after independent, family-specific inputs: if AP′ passes all negative controls, it is too weak; if an independently supported certificate separates a literature-known positive case from a stress case, it merits further formalization.

Towards AP′*: Provenance Defects*

The first strengthening should be read as a provenance scanner rather than as another positivity cone. Relative to a named test family, define a defect vector

$$D_{\mathrm{prov}}(\alpha) = (D_{\mathrm{slope}}, D_{\mathrm{Tate}}, D_{\mathrm{MT}}, D_{\mathrm{mon}}, D_{\mathrm{cycle}}),$$

where the entries record, respectively, slope compatibility at good reductions, explanation by the Tate/Galois fixed part, Mumford–Tate compatibility, monodromy or o-minimal persistence, and the residual class left after subtracting the known algebraic cycle span. A conditional fingerprint statement would say that, in a test family where the relevant Tate/standard-conjecture input and comparison functoriality are independently available, $D_{\mathrm{prov}}(\alpha) = 0$ places α in the known algebraic cycle span. This is a special-family diagnostic, not a global theorem.

Each component of D_{prov} must be recorded with status *verified*, *failed*, or *unknown*. Unknown components do not count as zero defects. This convention prevents the provenance scanner from becoming tautological: AP′ is passed only when all required components are independently verified in the chosen test family.

Towards AP″*: Conditional Orbit-Rigidity*

The five transfer slots suggest a prototype sharpening. Let $f\colon \mathcal{X}_S \to S$ be a spread-out of X over a smooth base S, let $Z \subset S$ be an irreducible algebraic component of the Hodge locus of α, let B denote the corresponding relative Chow or Hilbert component, and let $T \subset Z$ be a Zariski-dense subset at whose points α_t is algebraic relative to B. We call α *orbit-rigid on* (f, Z, B) if:

(i) *Properness.* The relative Chow/Hilbert components parametrizing the relevant cycle classes are proper over S.

(ii) *Density.* T is Zariski-dense in Z.

(iii) *Comparison functoriality.* The comparison isomorphisms between Betti, de Rham, ℓ-adic, and crystalline realizations are functorial on the relevant quotient; no new residual classes arise when lifting from a special to the generic fiber.

(iv) *Specialization rigidity.* Base change along $Z \to S$ does not create new residual cycle classes.

Lemma IX.28 (Conditional Orbit-Rigidity Template)**.** *Let $\alpha \in \mathrm{AbsHodge}^p(X)$ be orbit-rigid on (f, Z, B) in the sense above. Then the restriction α_η to the generic fiber $\mathcal{X}_\eta$ over the generic point η of Z lies in the image of the cycle class map* $\mathrm{cl}\colon \mathrm{CH}^p(\mathcal{X}_\eta)_{\mathbb{Q}} \to H^{2p}(\mathcal{X}_\eta, \mathbb{Q}(p))$*. If the base-point $0 \in S$ lies in the same controlled component, then α is algebraic. This is a checklist statement: the cycle-provenance, comparison, and specialization hypotheses are part of the input and are not verified by* AP*,* AP′*, or* AP‴ *alone.*

Table III. Multi-scale stability hierarchy of positivity gates beyond bilinear Hodge–Riemann relations.

Gate Level	Mathematical Definition / Invariant	Diagnostic Role	Decisive Benchmark / Boundary
AP (Bilinear)	$(-1)^p Q_{L^\sigma}(\alpha^\sigma, \alpha^\sigma) \geq 0,\ \forall \sigma, L$	Base HR consistency across $\mathrm{Aut}(\mathbb{C})$	Saturated: AP = AbsHodge (Rem. IX.2)
AP′ (Provenance)	$D_{\mathrm{prov}}(\alpha) = (D_{\mathrm{slope}}, D_{\mathrm{Tate}}, D_{\mathrm{MT}}, D_{\mathrm{mon}}, D_{\mathrm{cyc}}) = 0$	Multi-place crystalline & Galois fixed scan	Detects Frobenius slope anomalies (Sec. IX I)
AP″ (Orbit-Rigid)	Proper relative Hilbert B, dense $T \subset Z$, lifting rigidity	Bounded family propagation (Lem. IX.28)	Validated on Weil fourfolds of disc. 1
AP‴ (Anti-Advice)	$\mathrm{AP}'''_{\mathrm{unif}}(\alpha; R, C)$: Pre-registration, bounded advice, erosion	Excludes post-hoc tautological certificates	Separates genuine cycles from tailored rules
Reef_E (Multi-Scale)	$\mathrm{Reef}_E(\alpha; R, C, E)$: Multi-axis residues Res_{e_i}, $\mathrm{Ob}_{\mathrm{lift}} = 0$	Stratified boundary & lift obstruction clearance	Active frontier for non-abelian geometries

Proof idea. Properness of B *(i)* and Zariski-density of T *(ii)* imply via the valuative criterion that the closure of the algebraic locus in B contains the generic point η of Z. Comparison functoriality *(iii)* ensures that each algebraic class over $t \in T$ lifts consistently in all cohomological realisations. Specialisation rigidity *(iv)* prevents accumulation of transcendental residual classes at the limit. A complete proof requires the Cattani–Deligne–Kaplan principle for Hodge loci [8] together with a properness argument for the relative Hilbert scheme; both steps are conditional on the standard conjectures in any relevant characteristic-p reduction, or must be replaced by independent family-specific cycle-provenance input. □

Remark IX.29. Lemma IX.28 does not prove the Hodge Conjecture. It should be read as a conditional closure template, not as a new unconditional algebraicity theorem. Its force lies entirely in the independent verification of bounded relative Chow/Hilbert data, comparison functoriality, and specialization rigidity in a named test family. Under those independently checked hypotheses, algebraicity propagates from a Zariski-dense set to the generic point. Markman proves the algebraicity of Weil classes on abelian fourfolds of Weil type with complex multiplication by a quadratic imaginary field [22]; his Corollary 1.3 also establishes the Hodge Conjecture for abelian varieties of dimension at most five. These results identify known positive cases, but they do not verify this paper's AP″ certificate or its relative Chow/Hilbert, comparison, and specialization hypotheses. The $K3 \times K3$ perturbations remain a planned negative control, not an independently validated separation test.

Towards AP‴*: Uniformity and Anti-Advice*

The orbit-rigidity certificate still has a failure mode: the geometric family f, the Hodge-locus component Z, the relative Chow or Hilbert component B, or the dense set T could be chosen post hoc after inspecting the target class α. In that case the certificate may encode α rather than explain it. We therefore use

$$\mathrm{AP}'''_{\mathrm{unif}}(\alpha; R, C)$$

as an anti-tautology filter for an AP″ certificate C. It requires that the auxiliary data are generated from a pre-registered rule class R depending only on the ambient data (X, p) or on a fixed test-family ladder, with bounded advice, erosion stability under removal of an α-specific incidence condition, support on a neighboring family or Zariski-dense set, and failure on negative controls driven only by Hodge–Riemann form stability. This condition does not add a new global Hodge conjecture; it says when a special-family certificate is non-tautological.

Operationally, a certificate fails the AP‴ filter if any of the following checks fails:

1. *Pre-registration:* the rule class R was not fixed before the target class or does not follow from ambient data.
2. *Bounded advice:* the choices of f, Z, B, T carry enough information to reconstruct α.
3. *Erosion stability:* removing an α-specific incidence condition destroys the certificate.
4. *Neighbor-swarm:* the same rule does not support a neighboring family or a Zariski-dense set.
5. *Negative controls:* Voisin/$K3 \times K3$-type stress families pass merely because of Hodge–Riemann form stability.

Framework note. The Voisin torus is non-projective, so the ample-class-dependent Q_L test used here is undefined for it. The ledger records an out-of-framework case, not an AP2a failure or a computed negative sign.

Scale and Multi-Erosion Guardrails

The AP‴ filter still needs two practical guardrails before it can be used as a serious test-family protocol.

Scale specificity. A provenance detector should record a scale ledger

$$\text{scale_level}, \quad \text{detector_degree}, \quad \text{target_complexity_degree}$$

for the family under study. A detector that sees only a low-degree form signature is not a provenance bridge if the residual non-algebraic stress family has higher complexity. In that case the correct status is a specificity or capacity gap, not progress toward algebraicity. Operationally, the provenance matrix should add a residual entry

$$D_{\mathrm{scale}}(\alpha)$$

which is zero only when the detector resolves the same complexity level as the class it is meant to exclude. Operationally, this means $D_{\mathrm{scale}}(\alpha) = 0$ only if the recorded

Table IV. Controlled test-family diagnostic ledger and obstruction verification across geometric classes.

Geometric Family	**Cycle Mechanism**	AP′ **(Provenance)**	AP″ **(Rigidity)**	AP‴ **(Anti-Advice)**	**Evidence Status**	**Diagnostic Ledger**
CM / Weil Fourfolds ($d = 1$)	Weil classes [22]	AP′ unverified	Not tested here	Not tested here	Weil classes algebraic	No D_{prov} result
Abelian Varieties ($g \leq 5$)	Weil classes, Cor. 1.3 [22]	AP′ unverified	Not tested here	Not tested here	Hodge Conjecture known	No local AP certificate
OG6 Hyperkähler Families	Hyperkähler cycles	Monodromy/Cycle: PASS	Open / Unchecked	Pre-registered ledger	Structural test family	In progress
Char-p Reductions (Tate)	Crystalline Frobenius	Slope/Tate: PASS	Conditional lift	Reduction rule	Arithmetic benchmark	Model-dependent
Voisin Non-Projective Torus	No projective input	Outside framework	Not evaluated	Not evaluated	No AP verdict	No ample L; no Q_L test
$K3 \times K3$ Non-Tensor Slice	Perturbed non-Hodge	Residual $D_{\text{MT}} \neq 0$	Direct failure	Fails negative gate	Negative control	Fails Galois Hodge type

detector degree is at least the recorded target complexity degree. Otherwise the status is a specificity gap. In particular, Hodge–Riemann form signatures are degree-low diagnostics; by themselves they cannot explain K3×K3- or Voisin-type stress families as provenance data.

Multi-erosion provenance. For a pre-registered set of independent erosion axes

$$E = \{e_1, \ldots, e_m\},$$

write

$$\text{Reef}_E(\alpha; R, C, E)$$

for the following conditional extension of the anti-advice filter. The axes e_i are fixed from ambient family data, not from the coordinates of α; each boundary or residue image $\text{Res}_{e_i}(\alpha)$ is either explained by a fixed algebraic cycle span on the relevant stratum or recorded as a defect $D_{\text{res}}(e_i, \alpha)$; residues agree on overlaps in the Čech/Gysin/nearby-cycle sense; and the global lift obstruction

$$\text{Ob}_{\text{lift}}(\alpha; E) = 0.$$

vanishes. At least two independent pre-registered axes should be used; a single tailored axis is only a probe, not a reef certificate. This is not a new theorem. It is a future-work checklist: a special-family certificate should fail if the same class can be rebuilt by private erosion axes, incompatible boundary explanations, an unresolved Abel–Jacobi/normal-function/vanishing-cycle lift obstruction, or a negative control that passes under the same rule. Unknown residue, overlap, lift, or negative-control status is an open defect, not a pass.

Result Classification

This paper is a **boundary result**: it determines precisely what positivity-based methods can and cannot achieve for the Hodge conjecture.

X. CONCLUSION

We have proposed a reformulation of the Hodge Conjecture through the lens of positivity and No-Go obstructions. The key concept—*arithmetic positivity*—combines Hodge–Riemann positivity, absolute stability under Aut(ℂ), and Lefschetz compatibility into a single condition.

Table V. Comprehensive result and epistemic boundary classification of the arithmetic positivity program.

Result / Theorem	**Epistemic Status**	**Operational Role**
No-Go & Boundary Theorems:		
AP = AbsHodge (Thm. VI.1, Rem. IX.2)	Proven boundary	Saturates bilinear HR
APC hard direction ≡ Deligne's question	Exact equivalence	Scope limiter
Diagnostic Instruments:		
Galois–HR Spectrum (Def. VIII.1)	Definition	Visualizes Aut(ℂ) twist
Finite Pullback & Künneth (Prop. V.1, V.8)	Conditional / Partial	Subcone sign tests
Voisin GHR-Detection (Rem. IX.27)	Numerical / Diagnostic	Negative witness
Orbit-Rigidity Template AP″ (Lem. IX.28)	Conditional template	Propagation checklist
Uniform Anti-Advice Filter AP‴	Methodological filter	Excludes tautologies
Scale / Reef$_E$ Provenance (Sec. IX I)	Multi-axis guardrail	Structural framework
Open Frontiers (Deligne 1982):		
$\text{AbsHodge}^p(X) \stackrel{?}{=} \text{cl}(\text{CH}^p(X)_{\mathbb{Q}})$	Open (40+ years)	Core algebraicity
Stronger non-bilinear positivity	Active direction	Post-Deligne program

A central finding of this paper (Remark IX.2) is that the arithmetically positive cone coincides with Deligne's absolute Hodge classes: $\text{AP}^p(X) = \text{AbsHodge}^p(X)$. This means the Arithmetic Positivity Conjecture is equivalent to Deligne's question "Are all absolute Hodge classes algebraic?" The positivity conditions (AP1), (AP2b), and (AP3) are automatically satisfied by any absolute Hodge class, via the Hodge–Riemann bilinear relations.

The paper does not introduce a genuinely new conjecture; the saturation of Hodge–Riemann positivity for (p, p)-classes is a standard consequence of the HR relations. The paper's contributions are expository and systematic:

1. **A new perspective:** The No-Go formulation recasts the Hodge Conjecture from a construction problem to an impossibility statement.

2. **A visualization of the absoluteness test:** The Galois–Hodge–Riemann spectrum reformulates Deligne's absoluteness criterion in the language of Hodge–Riemann positivity, making explicit which Galois conjugation could witness non-algebraicity. It does not provide constraints beyond Deligne's absoluteness.

3. **Partial functoriality checks:** The paper records partial functoriality checks rather than a full functorial theory: finite pullback is conditional on the ample-cone comparison, primitive Künneth products provide product-polarization sign formulas only, and finite pushforward remains open.

The remaining challenge is Deligne's question: proving that absolute Hodge classes are algebraic. Future work

should seek stronger positivity conditions—beyond the Hodge–Riemann relations—that could further narrow the gap between absolute Hodge classes and algebraic classes.

To summarize the philosophical position of this paper: the Hodge Conjecture, viewed through the positivity lens, does not ask us to build algebraic cycles—it asks us to show that geometry has no room for anything else. The present work shows that Hodge–Riemann positivity alone cannot accomplish this; genuinely new positivity conditions are required.

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